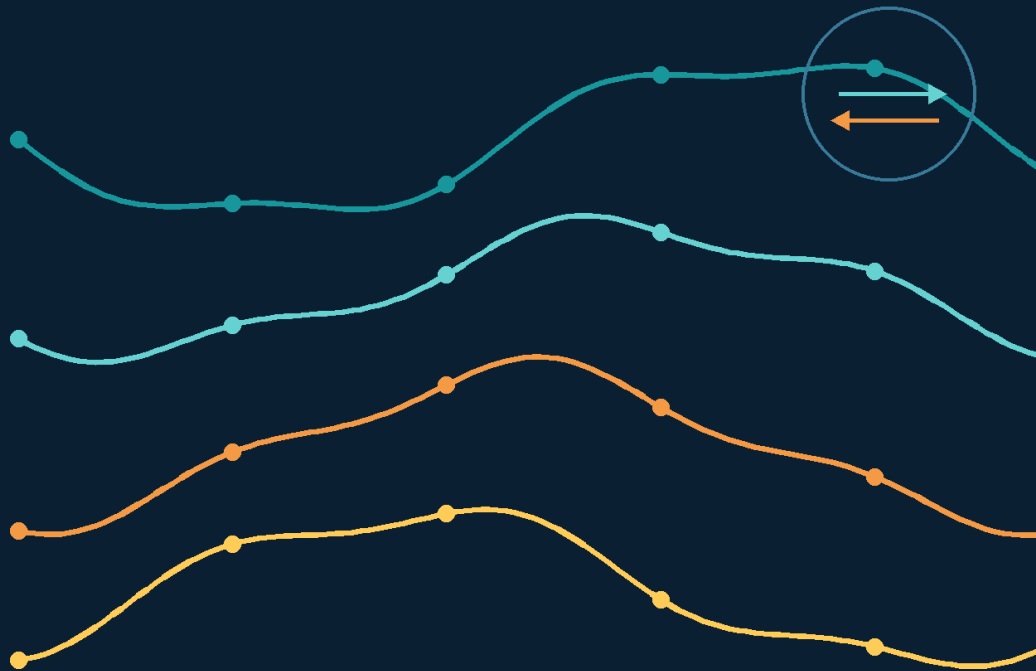


PHASE 5 • STANDALONE HANDBOOK



Displacement Distributions and Spatial Inversion Asymmetry

Time Irreversibility in Cell Motility

A cell-by-cell scientific, mathematical, biophysical, Python, and decision-making record of Phase 5

VALIDATED PHASE 5 SNAPSHOT

Exact-lag inputs 433 / 398 / 343 • raw + co-moving branches
FD histograms • 1D/2D KDE • KL inversion asymmetry
cell-level bootstrap • inversion-symmetrized null • sensitivity analysis

Prepared as a standalone continuation companion to the Phase 0–4 Master Handbook
Repository main and executed Notebook 05 inspected 5 August 2026

Document purpose, scope, and evidence standard

This standalone handbook documents Phase 5 only. It is intentionally separate from the validated Phase 0–4 Master Handbook so that the heavy probability-density and asymmetry work can be transferred cleanly to a future chat or collaborator without re-editing the earlier cumulative document.

DOCUMENT ROLE A scientific report, teaching text, notebook reconstruction, reproducibility record, decision log, and continuation handoff for Phase 5. It is not a manuscript and does not upgrade exploratory findings into publication-grade biological claims.

Evidence was reconciled against the current GitHub main branch, the executed Notebook 05 stored in the repository, the current README, and the outputs/screenshots generated during the Phase 5 session. The repository README inspected on 5 August 2026 states that Phase 5 is complete and Phase 6 is next; the executed notebook ends with “Phase 5 validation: PASSED.”

Evidence label	Meaning
REPOSITORY-VERIFIED	Present in GitHub main / current README or executed Notebook 05.
NOTEBOOK-OUTPUT VERIFIED	A value or figure stored as an executed notebook output.
SESSION-OUTPUT VERIFIED	Observed directly in screenshots/terminal output during the Phase 5 work.
PROJECT DECISION	A deliberate analysis/scope decision made during this phase.
INTERPRETATION BOUNDARY	A statement limiting what can be scientifically claimed.

EXACT STOPPING POINT Phase 5 is scientifically complete and validated. The next scientific task is Phase 6: ordered multi-step sequence analysis, where time reversal reverses both vector direction and temporal order.

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Appendix B. Numerical output ledger

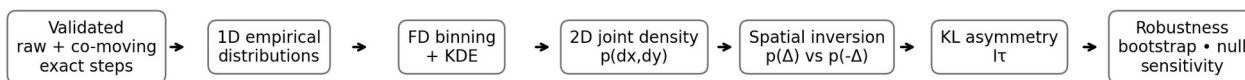
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1. Phase 5 handoff brief

Phase 5 analysis flow



Interpretation boundary: single-displacement spatial inversion asymmetry $\neq$ full sequence-level time irreversibility

Figure 1. Phase 5 workflow from validated raw/co-moving exact-lag steps to robustness-controlled spatial inversion asymmetry.

Item	Phase 5 verified state
Primary notebook	notebooks/05_displacement_density_MSC01.ipynb
Repository status	GitHub main inspected 5 Aug 2026; README reports Phase 5 completed and Phase 6 next.
Trajectory inputs	Raw and co-moving tables: 474 observations, 35 cells each; identical cell/frame/time structure.
Exact-lag steps	$\tau=1$: 433; $\tau=2$: 398; $\tau=4$: 343; 1174 steps per representation.
Combined analysis table	2348 rows (raw + co-moving) and 10 columns.
Main estimator	2D Gaussian KDE on a symmetric dx-dy grid; KL divergence between $p(\Delta)$ and $p(-\Delta)$.
Baseline I_τ (raw)	0.682156, 0.568221, 0.792825 for $\tau=1,2,4$.
Baseline I_τ (co-moving)	0.482956, 0.437690, 0.541848 for $\tau=1,2,4$.
Robustness	Bandwidth $0.5\times/1\times/2\times$, cell bootstrap, inversion-symmetrized null, constant- y sensitivity.
Final interpretation	Exploratory single-displacement spatial inversion asymmetry; not full temporal irreversibility or entropy production.
Next phase	Ordered multi-step sequence reversal: $(\Delta 1, \Delta 2, \Delta 3) \leftrightarrow (-\Delta 3, -\Delta 2, -\Delta 1)$.

LOCKED-IN SCIENTIFIC LANGUAGE Use “single-displacement spatial inversion asymmetry” or “displacement inversion asymmetry.” Do not rename the Phase 5 quantity “entropy production” or “full time irreversibility.”

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2. Scientific and mathematical foundations

Phase 5 changes the viewpoint from individual trajectory rows to probability distributions of displacement vectors. The central object is the exact-lag displacement vector $\Delta \mathbf{r}(t) = \mathbf{r}(t+\tau) - \mathbf{r}(t)$, evaluated separately for $\tau=1,2,4$ frames and for raw and co-moving coordinate representations.

Random variable and sample. Each observed dx or dy value is a sample (realization) from an empirical displacement variable. Pooling across cells and times creates an empirical ensemble. This is useful for density estimation, but the samples are not fully independent because multiple steps come from the same cell and multi-frame windows can overlap.

Probability density versus probability. A histogram with `density=True` is normalized by area, not by bar height. For a bin B_j with width w_j and count n_j , the estimated density is approximately $\hat{p}_j = n_j / (N w_j)$, while the probability mass in the bin is $\hat{p}_j w_j$.

Freedman–Diaconis rule. A data-driven histogram width $h = 2 \cdot \text{IQR} / n^{1/3}$ balances sample size and robust spread. It replaced the first pedagogical 30-bin choice as a robustness-oriented visualization rule, without pretending that histogram binning is a unique physical model.

Kernel density estimation (KDE). In one dimension, KDE replaces hard bins with a smooth sum of kernels. The bandwidth h controls the bias–variance trade-off: too small gives noisy peaks; too large washes out structure. In two dimensions, the same logic applies to the joint density $p(dx, dy)$.

Spatial inversion. A displacement vector $\Delta = (dx, dy)$ is mapped to $-\Delta = (-dx, -dy)$. This is not yet a full time-reversal operation because a one-step vector has no multi-step temporal ordering to reverse.

Phase 5 asymmetry metric. On a symmetric grid, the KDE estimate $p(\Delta)$ is compared with its inversion-reflected counterpart $p(-\Delta)$. The implemented quantity is the Kullback–Leibler divergence $\text{DKL}[p(\Delta) \parallel p(-\Delta)]$, evaluated numerically. It is non-negative and zero only when the estimated distribution is exactly inversion symmetric.

FINITE-SAMPLE WARNING Because KDEs estimated from finite samples are never perfectly symmetric, a positive KL value alone is not sufficient evidence. This is why Phase 5 included a symmetrized null reference and bandwidth sensitivity.

Cell-level bootstrap. Steps from a single cell are clustered. Resampling whole cells with replacement is therefore more defensible than treating all 433/398/343 steps as independent replicates. The bootstrap was used as exploratory uncertainty assessment, not as a definitive frequentist proof.

Biophysical interpretation. A nonzero one-step asymmetry can arise from directed migration, population translation, anisotropy, persistence, tracking/data structure, or genuinely non-equilibrium dynamics. Phase 4 already showed that a common population component exists, so Phase 5 deliberately compared raw and co-moving coordinates before making any stronger inference.

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3. Input data, repository state, and preflight

The Phase 5 notebook was started only after the exact-lag implementation was re-audited. The terminal preflight showed main synchronized with origin/main, src/steps.py containing merge/frame_end exact-frame logic rather than row-shift logic, and the scientific test suite reporting 9 passed. This preflight mattered because density estimation is meaningless if the random variable $\Delta\tau$ is mislabeled in time.

Input / invariant	Verified value
Raw trajectory table	MSC01_tracks_clean.csv; 474 rows × 5 canonical track columns; 35 cells.
Co-moving trajectory table	MSC01_tracks_comoving.csv; selected to the same 5 canonical track columns; 474 rows; 35 cells.
Structural identity	cell_id/frame/t_min structures exactly equal between raw and co-moving.
Lag set	TAUS = [1, 2, 4] → 20, 40, 80 minutes.
Exact raw steps	1174 total → {1:433, 2:398, 4:343}.
Exact co-moving steps	1174 total → {1:433, 2:398, 4:343}.
Combined tidy table	2348 rows × 10 columns.
Environment	Conda environment cell-irreversibility; tests 9 passed before Phase 5.

PREFLIGHT DECISION The project did not proceed until the exact-lag invariant was re-confirmed. This is an example of a problem worth stopping for: it changes the definition of the observable itself.

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4. Cells 1–8: data intake and exact-lag validation

This block establishes that Phase 5 is operating on the correct, matched, exact-lag data before any probability estimation.

Cell 1 — Define Phase 5 scope and stop premature irreversibility claims

Type / role	Markdown
Code or content summary	Introduced the session aim: load both coordinate representations, rebuild exact steps, validate counts, combine into tidy data, and inspect dx/dy before density estimation. Explicitly stated that no irreversibility metric was calculated yet.
Stored output	No computational output.
Interpretation / decision	Established a disciplined progression: understand the empirical random variables before estimating $p(\Delta)$ or interpreting asymmetry.
Concepts taught / reinforced	Notebook scientific narrative; scope control; interpretation boundaries.

Cell 2 — Import project helpers and define canonical paths

Type / role	Code
Code or content summary	Imported <code>_path/PROJECT_ROOT</code> , <code>pandas</code> , <code>validate_tracks</code> , <code>compute_steps_multi_tau</code> ; defined raw/co-moving CSV paths and <code>TAUS=[1,2,4]</code> ; printed path/existence checks.
Stored output	Both raw and co-moving files existed; lags printed as <code>[1, 2, 4]</code> .
Interpretation / decision	Confirmed notebook portability and that local ignored Phase 4 artifacts were available before analysis.
Concepts taught / reinforced	<code>pathlib</code> Path joining; imports; constants; reproducible project roots; <code>Boolean exists()</code> .

Cell 3 — Load and validate raw/co-moving trajectories

Type / role	Code
Code or content summary	Read both CSVs, selected canonical columns [<code>cell_id</code> , <code>frame</code> , <code>t_min</code> , <code>x_um</code> , <code>y_um</code>] from the co-moving file, called <code>validate_tracks</code> , reported shapes/cell counts, displayed heads.
Stored output	Raw shape (474,5); co-moving shape (474,5); 35 cells each. First five rows showed same identifiers/times with transformed coordinates.
Interpretation / decision	Confirmed the two branches are coordinate representations of the same observation set, not two biological datasets.
Concepts taught / reinforced	<code>DataFrame</code> ; <code>.loc</code> ; <code>.copy()</code> ; <code>nunique()</code> ; schema validation.

Cell 4 — Prove structural identity between representations

Type / role	Code
Code or content summary	Sorted and reset [<code>cell_id</code> , <code>frame</code> , <code>t_min</code>] for each table; compared with <code>DataFrame.equals()</code> ; asserted equality.
Stored output	Raw and co-moving trajectory structure identical: <code>True</code> .
Interpretation / decision	Made matched raw/co-moving comparison defensible: only spatial coordinates differ.
Concepts taught / reinforced	<code>DataFrame.equals</code> ; sorting; <code>reset_index</code> ; <code>assert</code> ; invariant testing.

Cell 5 — Recompute exact-lag steps from trajectories

Type / role	Code
Code or content summary	Called <code>compute_steps_multi_tau</code> separately on raw and co-moving tracks

	and copied results.
Stored output	Raw total steps: 1174; Co-moving total steps: 1174.
Interpretation / decision	Avoided dependence on a stale precomputed step CSV and reused the authoritative exact-frame function.
Concepts taught / reinforced	Function reuse; reproducible derivation; len().

Cell 6 – Validate the authoritative lag counts

Type / role	Code
Code or content summary	Built EXPECTED_COUNTS dict; groupby/size/to_dict on each branch; asserted equality.
Stored output	Raw counts {1:433,2:398,4:343}; Co-moving counts identical; "Phase 5 exact-lag input validation: PASSED."
Interpretation / decision	Locked the Phase 3/4 exact-lag baseline before any statistics.
Concepts taught / reinforced	Dictionary mapping; groupby; size; to_dict; assertions.

Cell 7 – Build a tidy raw + co-moving analysis table

Type / role	Code
Code or content summary	Added representation labels and concatenated the two step DataFrames with ignore_index=True; displayed core columns.
Stored output	Combined shape: (2348, 10). First rows were raw $\tau=1$ steps, e.g. cell 0 frame 10→11 dx=-5.091252, dy=3.172796.
Interpretation / decision	Created a uniform long-format table that supports systematic grouping by representation and lag.
Concepts taught / reinforced	pd.concat; tidy data; categorical labels; index rebuilding.

Cell 8 – Compute the first numerical distribution summary

Type / role	Code
Code or content summary	Grouped by representation and tau; aggregated n, mean, median, and standard deviation for dx/dy.
Stored output	Raw x means became more negative with lag (-0.449, -0.886, -1.687 μm). Co-moving x means were much nearer zero (-0.072, -0.133, -0.293). Standard deviations increased with lag.
Interpretation / decision	Confirmed Phase 4 visually/numerically: common translation explains a large part of raw x directional bias, but broad residual motion remains.
Concepts taught / reinforced	agg named aggregation; mean/median/std; lag-dependent spread; descriptive statistics.

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5. Cells 9–20: empirical distributions and numerical diagnostics

This block teaches probability-density visualization, then quantifies tails, skewness, and exact-zero structure before any smoothing model is trusted.

Cell 9 — Import numerical plotting tools

Type / role	Code
Code or content summary	Imported numpy and matplotlib.pyplot.
Stored output	No printed output.
Interpretation / decision	Prepared for controlled bin construction and visualization.
Concepts taught / reinforced	NumPy; Matplotlib.

Cell 10 — Isolate the first teaching example: $\tau=1$, dx

Type / role	Code
Code or content summary	Filtered steps_analysis for raw/co-moving $\tau=1$ dx using Boolean masks and .loc; printed sample sizes and ranges.
Stored output	433 observations in each branch. Raw dx range -12.8823001404 to 7.8000003047 μm ; co-moving -11.3519880929 to 7.8261344925 μm .
Interpretation / decision	Established equal sample sizes and motivated shared histogram support.
Concepts taught / reinforced	Boolean Series; .eq(); &; .loc; min/max.

Cell 11 — Create shared pedagogical histogram bins

Type / role	Code
Code or content summary	Concatenated raw/co-moving $\tau=1$ dx, found combined min/max, built 30 equal-width bins with np.linspace.
Stored output	Combined minimum -12.8823001404; maximum 7.8261344925; 31 edges $\rightarrow$ 30 bins.
Interpretation / decision	Ensured raw/co-moving plots use identical bin edges; 30 bins was explicitly pedagogical, not final.
Concepts taught / reinforced	np.linspace; bin edges vs bins; shared plotting geometry.

Cell 12 — Plot the first normalized raw histogram

Type / role	Code
Code or content summary	Used plt.hist(..., density=True) with the shared $\tau=1$ dx bins and a zero reference line.
Stored output	Histogram showed a single dominant central region slightly left of zero plus extended tails.
Interpretation / decision	Introduced probability density visually; highlighted that bar height is density and area integrates to ~ 1 .
Concepts taught / reinforced	Histogram density normalization; plotting; reference lines.

Cell 13 — Overlay raw and co-moving $\tau=1$ dx histograms

Type / role	Code
Code or content summary	Plotted both branches on identical bins with transparency and legend.
Stored output	Co-moving mass shifted toward zero; raw distribution remained more negative.

Interpretation / decision	Visual confirmation that common motion affects the center of the x distribution without eliminating individual-cell variability.
Concepts taught / reinforced	Fair overlay comparisons; shared bin edges; alpha/legend.

Cell 14 – Define reusable displacement selection

Type / role	Code
Code or content summary	Created <code>get_displacement_series(data, representation, tau_frames, component)</code> with dx/dy mapping and validation.
Stored output	No direct output.
Interpretation / decision	Replaced repetitive filtering with one auditable reusable function.
Concepts taught / reinforced	Functions; parameters; return values; type hints; dictionaries; defensive <code>ValueError</code> .

Cell 15 – Define shared-bin helper

Type / role	Code
Code or content summary	Created <code>make_shared_bins(first, second, n_bins=30)</code> , combining samples and returning shared <code>np.linspace</code> edges.
Stored output	No direct output.
Interpretation / decision	Separated data selection from histogram geometry; introduced default arguments.
Concepts taught / reinforced	Default parameters; <code>np.ndarray</code> return type; single-responsibility functions.

Cell 16 – Define reusable histogram plotting function

Type / role	Code
Code or content summary	Created <code>plot_raw_vs_comoving_histogram()</code> that calls the selection and shared-bin helpers and plots normalized overlays.
Stored output	No direct output.
Interpretation / decision	Applied DRY and single-responsibility principles; made six comparisons consistent.
Concepts taught / reinforced	Function composition; side effects; <code>-> None</code> ; reusable plotting.

Cell 17 – Generate all six raw/co-moving histogram comparisons

Type / role	Code
Code or content summary	Called the plotting function for dx and dy at $\tau=1,2,4$.
Stored output	Six figures. x raw distributions remained more negative than co-moving; widths increased with lag. dy branches overlapped more strongly and showed a sharp central peak plus long tails.
Interpretation / decision	Moved from one teaching example to the complete 1D empirical overview; prompted numerical tail/zero diagnostics before KDE.
Concepts taught / reinforced	Loops/function reuse conceptually; comparative visualization.

Cell 18 – Build full numerical distribution diagnostics

Type / role	Code
Code or content summary	Nested loops over representation $\times$ lag $\times$ component; computed min/max, 1/5/25/50/75/95/99% quantiles, IQR, skewness, exact-zero count/fraction; accumulated list of dictionaries.
Stored output	12-row diagnostic table covering all 12 distributions.

Interpretation / decision	Converted visual impressions into quantitative diagnostics and exposed unusually large exact-zero dy mass and negative dy skew.
Concepts taught / reinforced	Nested loops; list of dictionaries; tuple unpacking; quantiles; IQR; skewness; Boolean mean.

Cell 19 – Display a compact diagnostic table

Type / role	Code
Code or content summary	Selected q05, median, q95, IQR, skewness, and exact-zero fraction from the full diagnostics.
Stored output	Notable examples: raw $\tau 1$ dy zero fraction 0.200924 and skewness -0.695; raw $\tau 4$ dy skewness -0.921; co-moving $\tau 4$ dy skewness -1.035.
Interpretation / decision	Confirmed that the y zero spike and left-tail asymmetry were real numerical features, not only histogram impressions.
Concepts taught / reinforced	Feature selection; robust summaries.

Cell 20 – Locate the largest raw $\tau=4$ dy displacements

Type / role	Code
Code or content summary	Defined show_extreme_steps(); ranked by absolute displacement with .abs() and .nlargest(); displayed top 10.
Stored output	Seven of the ten largest dy rows came from cell 10; the largest was -34.010622 μm (frame 21→25). Cell 7 contributed two large negative rows.
Interpretation / decision	Revealed that the long negative tail was strongly influenced by repeated overlapping windows from a small number of trajectories; no values were removed.
Concepts taught / reinforced	Absolute value; nlargest; outlier $\neq$ bad data; cluster dependence.

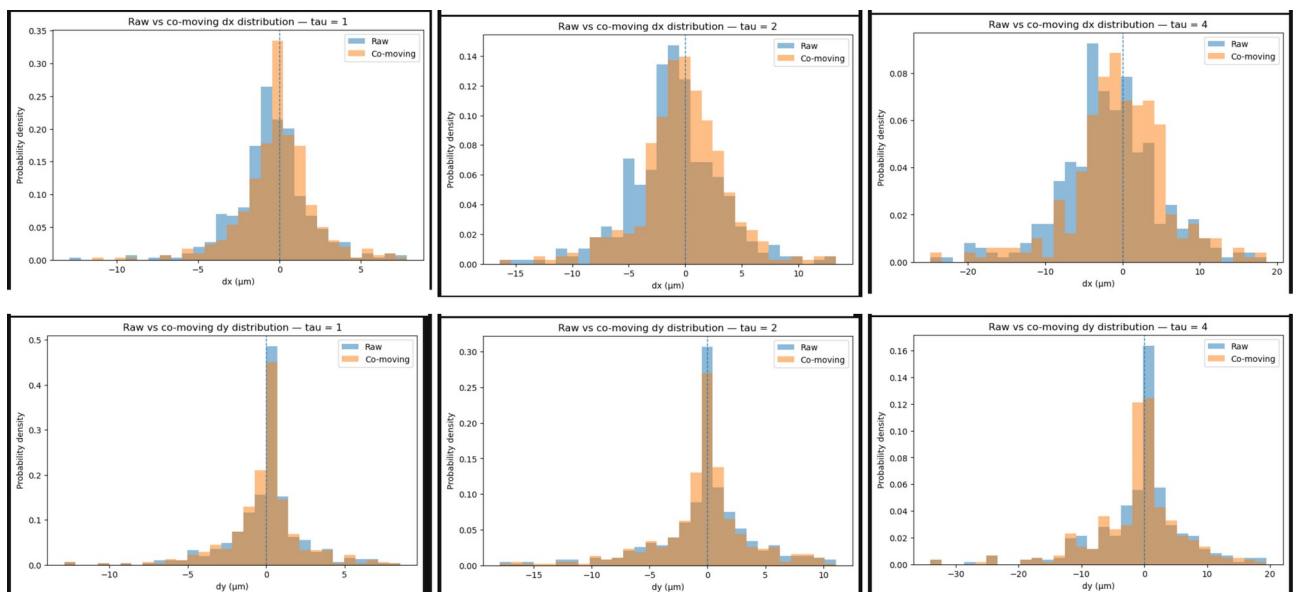


Figure 2. Stored raw-versus-co-moving histogram outputs for dx and dy at $\tau=1,2,4$. The broader distributions at larger lag and the strong dy peak near zero motivated quantitative diagnostics.

repr	τ	comp	q05	median	q95	IQR	skew	zero frac
raw	1	dx	-4.149	-0.483	3.319	2.143	-0.277	0.039
raw	1	dy	-4.522	0	3.801	1.790	-0.695	0.201
raw	2	dx	-7.512	-1.043	5.903	4.070	-0.058	0.018
raw	2	dy	-7.612	0	6.885	3.093	-0.777	0.186
raw	4	dx	-12.097	-1.747	9.219	7.064	-0.171	0.020

repr	τ	comp	q05	median	q95	IQR	skew	zero frac
raw	4	dy	-12.596	0	10.062	5.689	-0.921	0.166
co	1	dx	-3.764	0	3.334	1.984	-0.449	0.058
co	1	dy	-4.194	0	3.733	1.588	-0.716	0.132
co	2	dx	-6.702	-0.143	5.824	3.877	-0.242	0
co	2	dy	-7.935	0	6.154	2.817	-0.844	0.028
co	4	dx	-11.242	-0.315	9.208	6.880	-0.448	0
co	4	dy	-12.552	-0.271	9.202	5.047	-1.035	0

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6. Cells 21–30: diagnostic detour, data-quality findings, and scope decision

This block follows the Cell 10 and constant-y findings far enough to understand their influence, then deliberately stops before publication-grade forensic validation.

Cell 21 — Inspect Cell 10 raw trajectory table

Type / role	Code
Code or content summary	Filtered raw_tracks to cell_id=10, sorted by frame, displayed the track.
Stored output	Cell 10 trajectory values were displayed across its observed frames.
Interpretation / decision	Prepared to distinguish sustained biological motion from an isolated tracking jump.
Concepts taught / reinforced	Filtering a single entity; trajectory inspection.

Cell 22 — Plot Cell 10 raw y-position over time

Type / role	Code
Code or content summary	Line plot of frame versus y_um.
Stored output	Smooth descent from ~249 μm to ~173 μm followed by recovery toward ~217 μm .
Interpretation / decision	The large $\tau=4$ negative values were compatible with a sustained directional episode rather than a single obvious discontinuity.
Concepts taught / reinforced	Position-vs-time visualization; persistence.

Cell 23 — Inspect Cell 10 exact $\tau=1$ steps and 2D magnitudes

Type / role	Code
Code or content summary	Selected cell 10 $\tau=1$ steps; computed step_magnitude_um=np.hypot(dx,dy); displayed frame-wise values.
Stored output	Consecutive step table displayed.
Interpretation / decision	Connected large multi-frame displacements to sums of overlapping one-frame motions.
Concepts taught / reinforced	Euclidean magnitude; np.hypot; overlapping windows.

Cell 24 — Plot Cell 10 one-frame dy sequence

Type / role	Code
Code or content summary	Plotted $\tau=1$ dy versus starting frame with zero reference.
Stored output	Several consecutive negative steps, then a later sign change to positive motion.
Interpretation / decision	Supported the interpretation of persistent episodes and demonstrated why overlapping $\tau=4$ windows are correlated.
Concepts taught / reinforced	Temporal sequence visualization; serial correlation intuition.

Cell 25 — Count exact-zero dy by cell

Type / role	Code
Code or content summary	Grouped raw $\tau=1$ steps by cell; counted n_steps and n_zero_dy using a lambda; computed fraction and sorted descending.
Stored output	Cells 2,12,14,30,31,26,21 had 100% zero $\tau=1$ dy; cells 13/11/28 also had high fractions.

Interpretation / decision	Shown that the zero spike was concentrated in a subset of tracks rather than uniformly distributed across the dataset.
Concepts taught / reinforced	groupby/agg; lambda; proportions; per-cluster diagnostics.

Cell 26 – Separate zero-dy from full zero displacement

Type / role	Code
Code or content summary	Counted exact zero dx, exact zero dy, and simultaneous zero in both coordinates.
Stored output	Exact zero dx=17; exact zero dy=87; exact zero in both=13.
Interpretation / decision	74 $\tau=1$ steps had dy=0 while dx $\neq$ 0; the spike was not simply “cells did not move.” This motivated one more integrity check.
Concepts taught / reinforced	Boolean conjunction; exact-value diagnostics.

Cell 27 – Quantify large $\tau=1$ 2D step magnitudes

Type / role	Code
Code or content summary	Computed magnitude for all raw $\tau=1$ steps, printed maximum and count $>8\ \mu\text{m}$, displayed top 10.
Stored output	Maximum 12.9816169932 μm ; 9 $\tau=1$ steps $>8\ \mu\text{m}$. Large steps occurred across multiple cells (6,27,10,18,3,17,21,28,7).
Interpretation / decision	Large steps were not a single-cell anomaly. They were retained; the analysis did not escalate into exhaustive tracking validation.
Concepts taught / reinforced	Magnitude thresholds; population-wide anomaly check.

Cell 28 – Inspect Cell 21, a constant-y example

Type / role	Code
Code or content summary	Displayed cell 21 coordinates and counted unique x/y positions.
Stored output	Unique x positions=26; unique y positions=1; y values=[0.].
Interpretation / decision	Confirmed a true data structure: the stored y coordinate was exactly constant for a long track while x changed.
Concepts taught / reinforced	nunique; unique; coordinate-level provenance clue.

Cell 29 – Summarize all 100%-zero-dy tracks

Type / role	Code
Code or content summary	Collected all-zero-dy cell IDs into a list; looped over tracks to compute positions, unique x/y counts, x/y ranges.
Stored output	Cells [2,12,14,30,31,26,21]. Every listed track had n_unique_y=1 and y_range=0; cell 21 x_range $\approx$ 49.2 μm .
Interpretation / decision	Established a real data-quality limitation affecting a substantial subset, but not yet its cause.
Concepts taught / reinforced	tolist; loops; per-track summary DataFrame.

Cell 30 – Visualize Cell 21 x and y coordinates over time

Type / role	Code
Code or content summary	Plotted x_um and y_um against frame.
Stored output	x changed over time while y stayed at 0 for the entire track.
Interpretation / decision	This was the point where the project reconsidered scope: the issue was documented, but publication-grade forensic provenance work was judged disproportionate to the training-project goal.

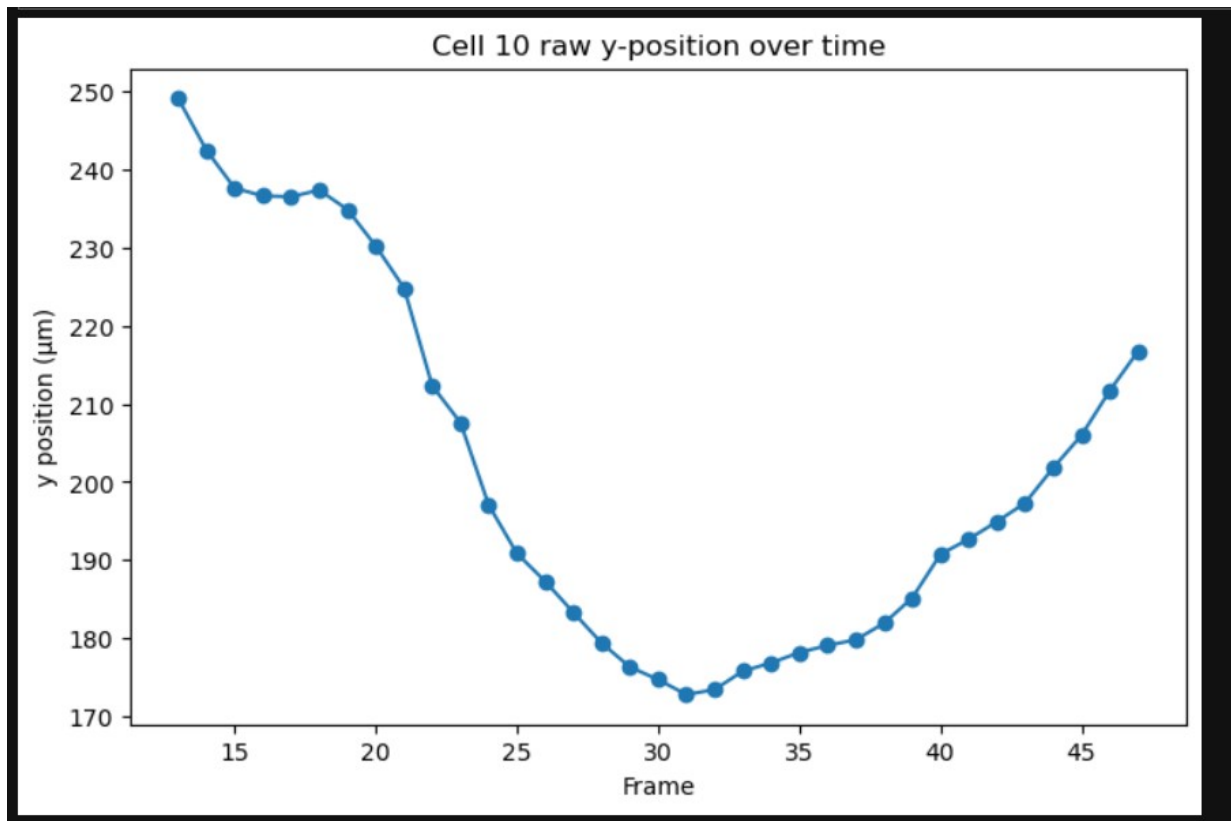


Figure 3. Cell 10 raw y-position over time. The long negative $\tau=4$ tail was traced to a sustained downward episode followed by recovery, not a single isolated jump.

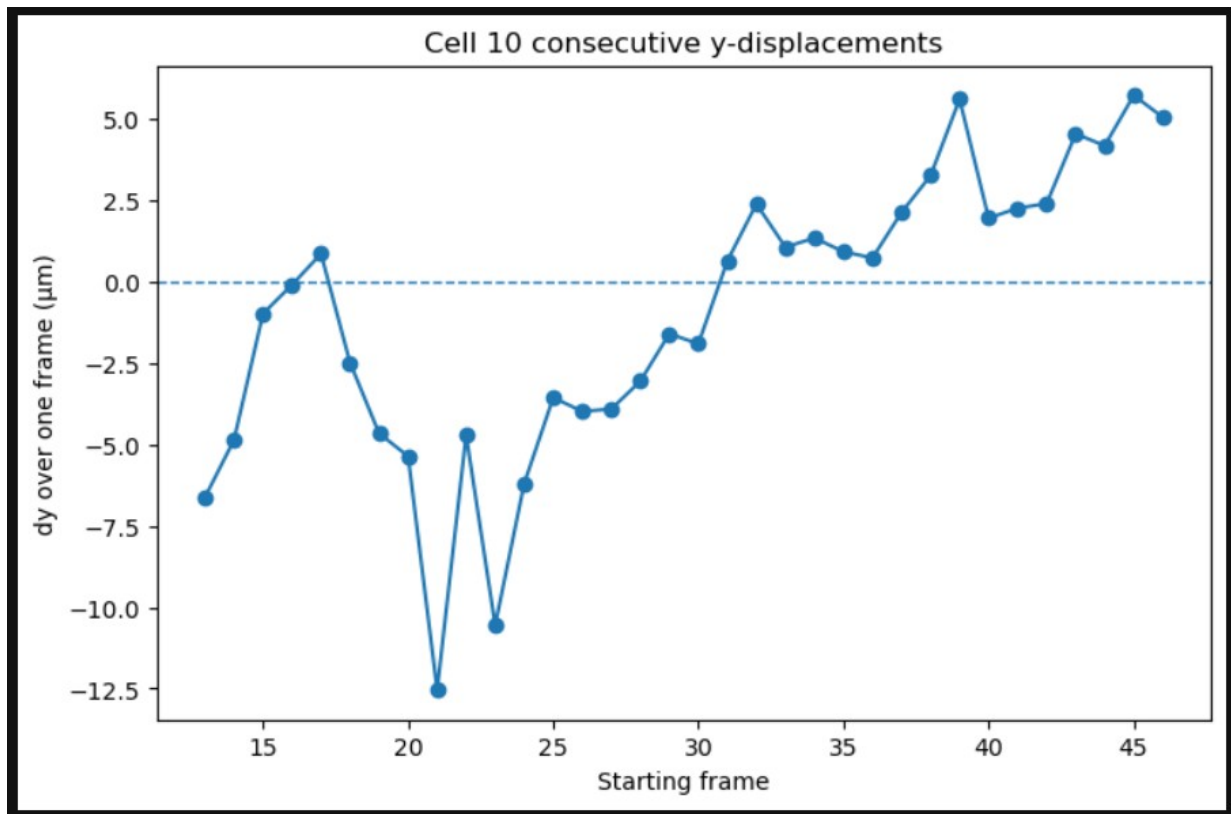


Figure 4. Cell 10 consecutive $\tau=1$ y-displacements. Overlapping multi-frame windows therefore reuse the same persistent episode and are not independent events.

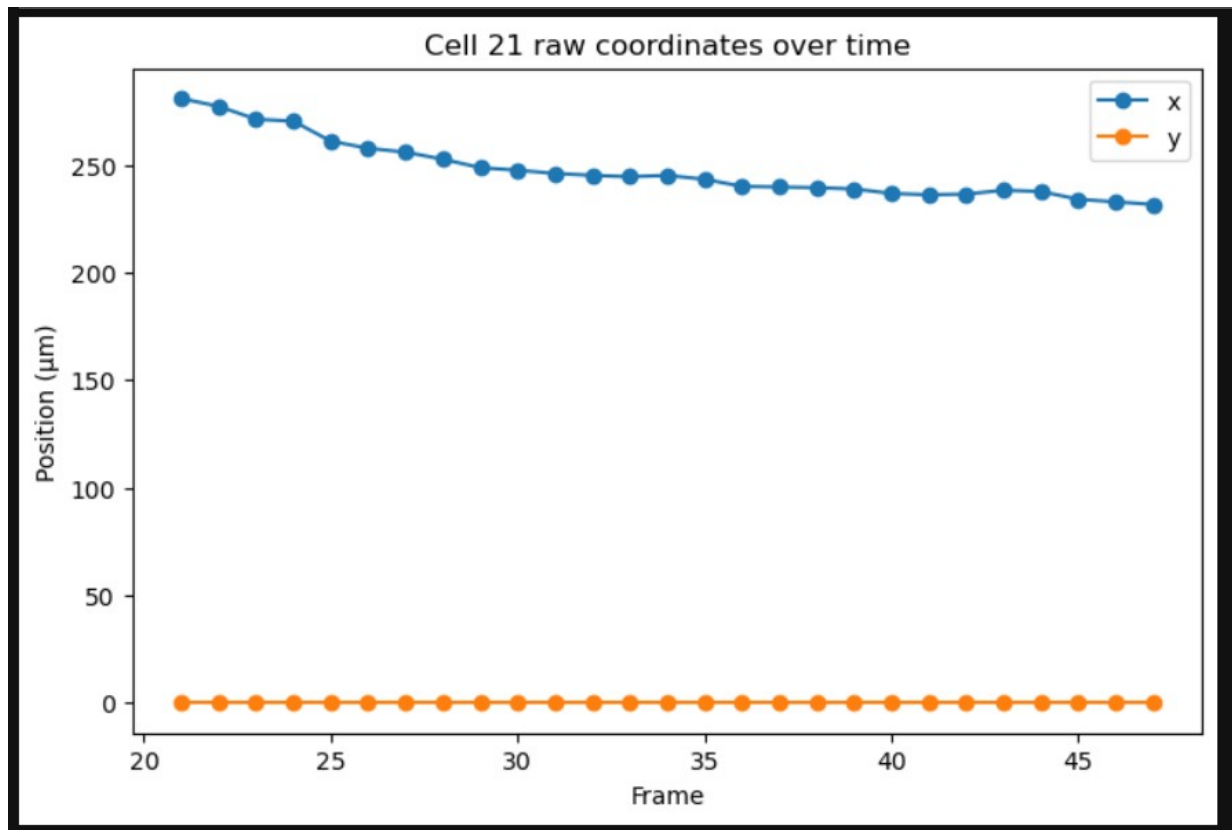


Figure 5. Cell 21 raw coordinates over time: x varies while y remains exactly zero, illustrating the constant- y track limitation.

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7. Cells 31–42: documented limitations, Freedman–Diaconis binning, and 1D KDE

This block records limitations formally, moves from pedagogical fixed bins to data-driven FD bins, and introduces smooth 1D KDE diagnostics.

Cell 31 — Record the Phase 5 data-quality policy

Type / role	Markdown
Code or content summary	Documented large overlapping displacements, constant-y tracks, decision to retain validated tracks in the primary analysis, and plan for a simple exclusion sensitivity analysis later.
Stored output	No computational output.
Interpretation / decision	Formalized the new project mindset: serious exploratory science, transparent limitations, and sensitivity analysis instead of endless forensic investigation unless validity is fundamentally threatened.
Concepts taught / reinforced	Scientific writing; limitation statements; stop rules.

Cell 32 — Define Freedman–Diaconis bin-width function

Type / role	Code
Code or content summary	Computed n , q_{25}/q_{75} , IQR, $h=2 \cdot \text{IQR}/\text{cuberoor}(n)$; raised errors for $n < 2$ or $\text{IQR} = 0$.
Stored output	No direct output.
Interpretation / decision	Replaced arbitrary bin count with a data-driven visualization rule.
Concepts taught / reinforced	Freedman–Diaconis rule; <code>np.cbrt</code> ; defensive programming.

Cell 33 — Convert FD width into shared raw/co-moving bin edges

Type / role	Code
Code or content summary	Combined two samples, computed FD width, range, <code>ceil(range/width)</code> , and <code>np.linspace</code> edges.
Stored output	No direct output.
Interpretation / decision	Preserved fair raw/co-moving comparison while making binning data-driven.
Concepts taught / reinforced	<code>np.ceil</code> ; width-to-bin-count conversion.

Cell 34 — Audit FD recommendations across all 12 distributions

Type / role	Code
Code or content summary	Nested loops calculated FD width and implied bin count for each representation/lag/component.
Stored output	Examples: raw τ_1 dx width 0.566533 (37 bins), raw τ_1 dy 0.473129 (45); co-moving τ_1 dy 0.419863 (51); τ_4 dx ≈ 22 bins in both branches.
Interpretation / decision	Showed the initial 30-bin choice was not absurd but was not universally optimal; justified FD plots as the more principled histogram view.
Concepts taught / reinforced	Systematic parameter audit; data-driven visualization.

Cell 35 — Define FD histogram comparison plotter

Type / role	Code
Code or content summary	Created <code>plot_fd_histogram_comparison()</code> using <code>get_displacement_series</code> and <code>make_fd_bins</code> .

Stored output	No direct output.
Interpretation / decision	Encapsulated the data-driven plotting policy.
Concepts taught / reinforced	Function reuse; separation of estimator/visualization.

Cell 36 – Generate six FD histogram comparisons

Type / role	Code
Code or content summary	Looped over τ and component to plot raw vs co-moving with shared FD bins.
Stored output	Six FD-based comparison figures generated.
Interpretation / decision	Confirmed qualitative histogram features were not artifacts of choosing exactly 30 bins.
Concepts taught / reinforced	Loop automation; binning robustness.

Cell 37 – Import scikit-learn KDE estimator

Type / role	Code
Code or content summary	Imported KernelDensity from sklearn.neighbors.
Stored output	No direct output.
Interpretation / decision	Moved from discontinuous histograms to smooth exploratory density estimation.
Concepts taught / reinforced	Library reuse; sklearn API.

Cell 38 – Define robust 1D Silverman-type bandwidth

Type / role	Code
Code or content summary	Computed sample std, IQR, robust scale= $\min(\text{std}, \text{IQR}/1.34)$, then $0.9 \cdot \text{scale} \cdot n^{-(1/5)}$.
Stored output	No direct output.
Interpretation / decision	Provided a reproducible baseline bandwidth without heavy bandwidth optimization.
Concepts taught / reinforced	ddof=1; robust scale; rule-of-thumb bandwidth.

Cell 39 – Fit/evaluate 1D Gaussian KDE

Type / role	Code
Code or content summary	Converted Series to (n,1), fit KernelDensity, evaluated log-density on a grid, exponentiated to density.
Stored output	No direct output.
Interpretation / decision	Introduced numerical log-density handling and array-shape conventions required by sklearn.
Concepts taught / reinforced	reshape(-1,1); log-space numerical stability; np.exp.

Cell 40 – Define raw/co-moving KDE comparison plotter

Type / role	Code
Code or content summary	Built common evaluation grid; selected separate Silverman bandwidths for each branch; evaluated and plotted densities; printed bandwidths.
Stored output	No direct output until called.
Interpretation / decision	Created the reusable 1D KDE diagnostic pipeline.
Concepts taught / reinforced	Function composition; evaluation grids; estimator visualization.

Cell 41 – Run first KDE checks at $\tau=1$

Type / role	Code
Code or content summary	Called KDE comparison for dx and dy.
Stored output	dx bandwidths: raw 0.4274292795, co-moving 0.3956570488 μm . dy bandwidths: raw 0.3569593675, co-moving 0.3167722591 μm . dx co-moving peak moved toward zero; dy branches nearly overlapped with a sharp zero-centered peak.
Interpretation / decision	KDE reproduced histogram conclusions while making the central structure easier to compare. The sharp dy peak remained documented as partly driven by constant-y tracks.
Concepts taught / reinforced	Bandwidth interpretation; smooth density diagnostics.

Cell 42 – Generate the complete set of 1D KDE comparisons

Type / role	Code
Code or content summary	Nested loop called plot_kde_comparison for all $\tau \times \{dx, dy\}$.
Stored output	Six KDE figures generated.
Interpretation / decision	Closed the 1D density stage and moved to the actual 2D displacement vector distribution.
Concepts taught / reinforced	Looped estimator workflow.

representation	τ	component	FD width (μm)	bins
raw	1	dx	0.566533	37
raw	1	dy	0.473129	45
raw	2	dx	1.106512	27
raw	2	dy	0.841010	35
raw	4	dx	2.018363	22
raw	4	dy	1.625413	33
co-moving	1	dx	0.524420	37
co-moving	1	dy	0.419863	51
co-moving	2	dx	1.054072	29
co-moving	2	dy	0.765841	37
co-moving	4	dx	1.965579	22
co-moving	4	dy	1.442101	37

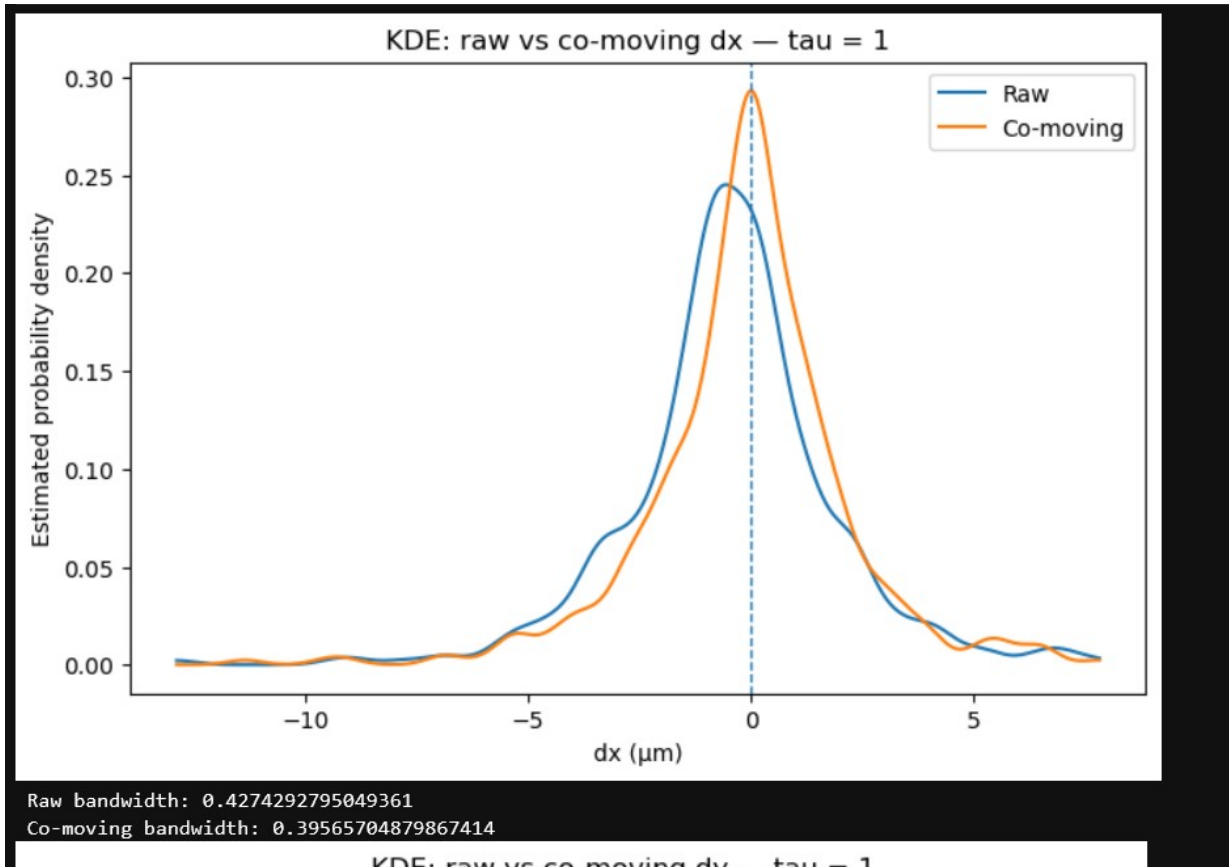


Figure 6. $\tau=1$ dx KDE: co-moving density shifts the central peak toward zero relative to raw; raw and co-moving bandwidths were 0.4274 and 0.3957 μm .

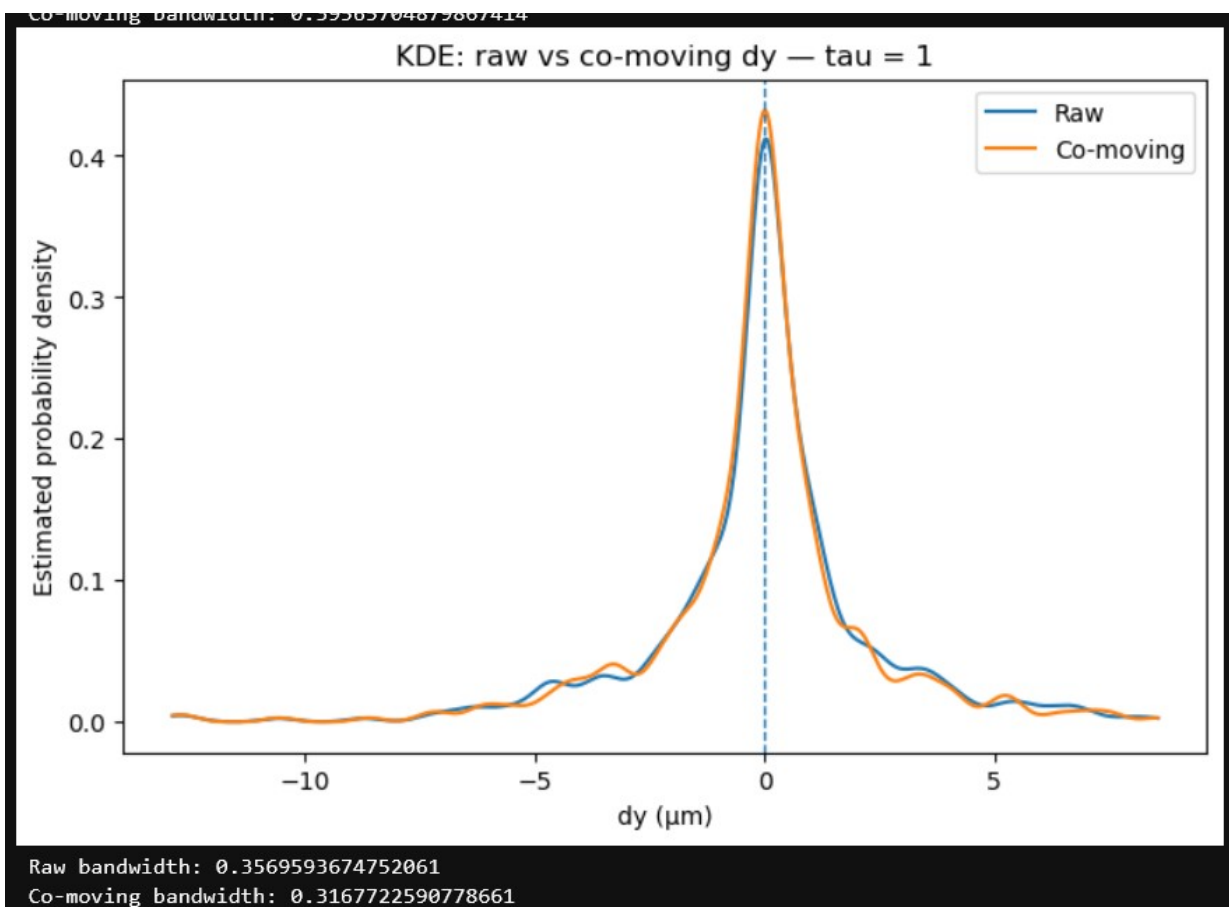


Figure 7. $\tau=1$ dy KDE: branches largely overlap and retain the sharp central peak; bandwidths were 0.3570 and 0.3168 μm .

8. Cells 43–51: 2D density, spatial inversion, and the baseline asymmetry metric

This block defines the full vector probability density, constructs the symmetric grid, implements $p(\Delta)$ vs $p(-\Delta)$, and produces the first central Phase 5 asymmetry values.

Cell 43 – Define the 2D joint-density and inversion problem

Type / role	Markdown
Code or content summary	Stated $\Delta r = (\Delta x, \Delta y)$, $p(\Delta x, \Delta y)$, spatial inversion $-\Delta r = (-\Delta x, -\Delta y)$, and the interpretation boundary.
Stored output	No computational output.
Interpretation / decision	Shifted from marginal 1D projections to the full vector distribution required for $p(\Delta)$ vs $p(-\Delta)$.
Concepts taught / reinforced	Joint distributions; vector inversion; interpretation boundary.

Cell 44 – Extract 2D displacement point arrays

Type / role	Code
Code or content summary	Defined <code>get_displacement_points()</code> ; selected <code>dx/dy</code> and converted to NumPy. Printed <code>raw</code> shape and first five points.
Stored output	Shape (433,2); first point [-5.09125217, 3.17279647], then [-3.72019571, 2.15500559], etc.
Interpretation / decision	Confirmed the correct <code>n_samples × n_features</code> structure for 2D KDE.
Concepts taught / reinforced	NumPy arrays; <code>to_numpy</code> ; 2D feature matrices.

Cell 45 – Define robust characteristic scale per coordinate

Type / role	Code
Code or content summary	Converted array to float, computed std and IQR/1.34, filtered invalid/nonpositive scales, returned minimum.
Stored output	No direct output.
Interpretation / decision	Built a conservative scale estimator less dominated by extreme tails.
Concepts taught / reinforced	<code>np.asarray</code> ; list comprehension; <code>np.isfinite</code> ; robust scale.

Cell 46 – Define a shared isotropic 2D KDE bandwidth

Type / role	Code
Code or content summary	Stacked raw/co-moving points; estimated x/y robust scales; geometric mean characteristic scale; $n^{-(1/6)}$ 2D scaling.
Stored output	No direct output.
Interpretation / decision	Used one shared bandwidth per lag so raw/co-moving smoothing is directly comparable. This was an intentional simplification instead of a full bandwidth matrix.
Concepts taught / reinforced	<code>np.vstack</code> ; dimensional scaling $n^{-(1/(d+4))}$; geometric mean.

Cell 47 – Construct symmetric 2D evaluation grids

Type / role	Code
Code or content summary	Computed max absolute x/y over combined data plus 3 bandwidth margins; built grids from <code>-limit</code> to <code>+limit</code> .
Stored output	No direct output.

Interpretation / decision	Guaranteed that each grid location Δ has a matching $-\Delta$ location and reduced tail truncation.
Concepts taught / reinforced	Symmetric numerical domains; finite-grid margining.

Cell 48 – Evaluate a 2D Gaussian KDE on the grid

Type / role	Code
Code or content summary	Used np.meshgrid, column_stack of flattened grid points, KernelDensity fit/score, exp, reshape to grid.
Stored output	No direct output.
Interpretation / decision	Produced a numerical density surface $\hat{p}(dx,dy)$.
Concepts taught / reinforced	meshgrid; ravel; column_stack; 2D KDE.

Cell 49 – Define the inversion-asymmetry index

Type / role	Code
Code or content summary	Computed grid-cell area and mass, normalized density, created $p_{\text{reversed}}=p[:,::-1,::-1]$, added tiny epsilon, renormalized, formed log ratio and integral $\sum p \log(p/q)$ area.
Stored output	No direct output.
Interpretation / decision	Implemented KL divergence between estimated density and its spatial inversion. Epsilon was numerical regularization, not a biological parameter.
Concepts taught / reinforced	Numerical integration; array reversal; KL divergence; epsilon regularization; nats.

Cell 50 – Compute baseline asymmetry for all lags and branches

Type / role	Code
Code or content summary	For each τ : extracted raw/co-moving points, computed shared bandwidth and symmetric grid, evaluated KDE and KL asymmetry, stored n, bandwidth, grid mass, l.
Stored output	τ_1 raw 0.682156 vs co 0.482956; τ_2 0.568221 vs 0.437690; τ_4 0.792825 vs 0.541848 nats. Grid masses 0.999996–0.999999.
Interpretation / decision	First central Phase 5 result: raw > co-moving at all lags, but co-moving asymmetry remained positive. Near-unit grid mass validated support coverage.
Concepts taught / reinforced	Tidy result records; 2D density comparison; KL interpretation.

Cell 51 – Visualize raw $\tau=1$ 2D displacement KDE

Type / role	Code
Code or content summary	Recomputed τ_1 raw/co common bandwidth/grid; plotted raw density with contourf and zero axes.
Stored output	2D density concentrated near origin with visible non-circular extensions, especially toward negative directions.
Interpretation / decision	Made the abstract joint density tangible; figure supported but did not independently prove asymmetry.
Concepts taught / reinforced	Contour density visualization; joint spatial structure.

repr	τ	n	bandwidth μm	grid mass	l (nats)
raw	1	433	0.515968	0.999998	0.682156
co-moving	1	433	0.515968	0.999999	0.482956
raw	2	398	0.955681	0.999998	0.568221

repr	τ	n	bandwidth μm	grid mass	l (nats)
co-moving	2	398	0.955681	0.999997	0.437690
raw	4	343	1.755631	0.999996	0.792825
co-moving	4	343	1.755631	0.999998	0.541848

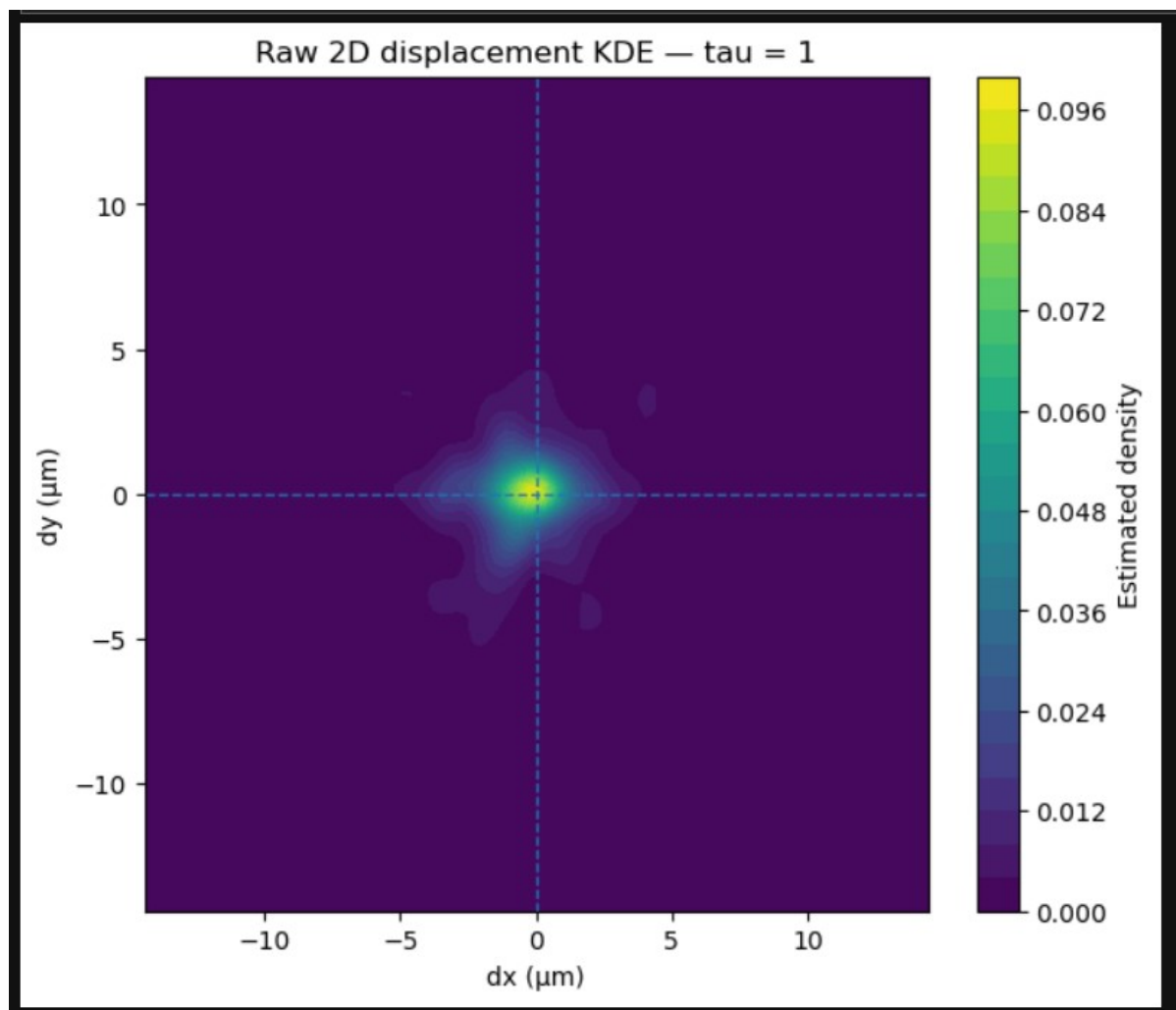


Figure 8. Stored raw 2D displacement KDE at $\tau=1$. The density is concentrated near the origin but is not visually circular/inversion-symmetric.

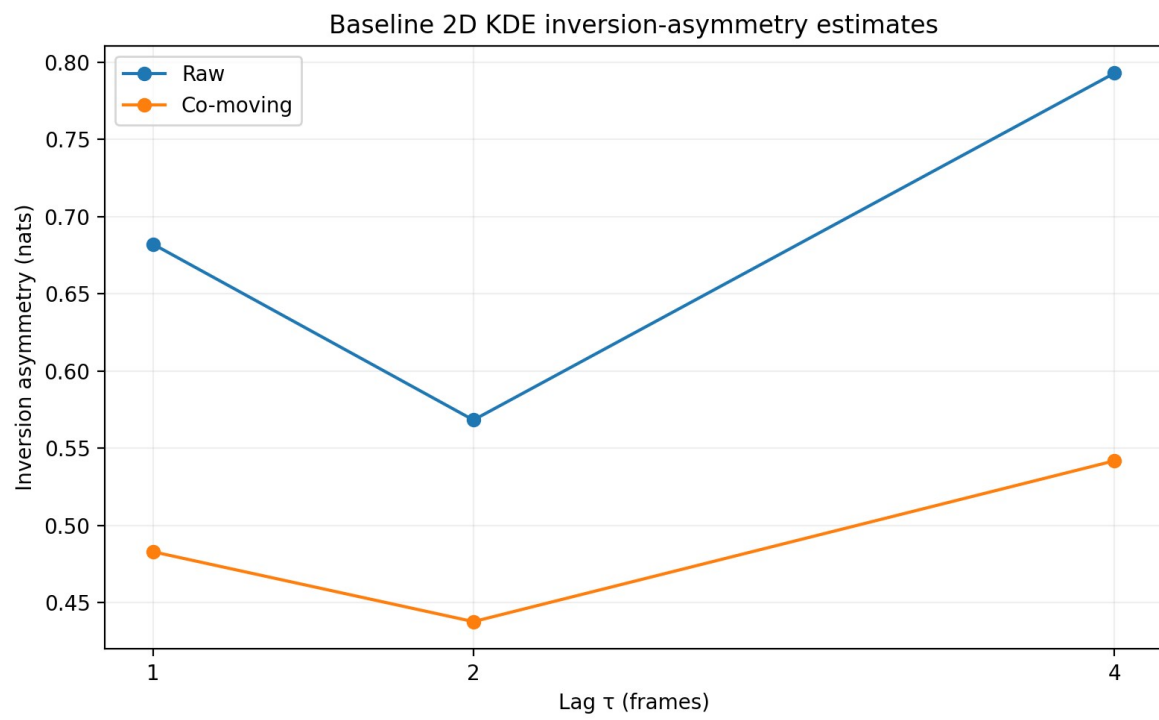


Figure 9. Baseline inversion-asymmetry estimates. Raw exceeds co-moving at all three lags.

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9. Cells 52–55: bandwidth robustness

This block tests whether the main raw-versus-co-moving ordering is an artifact of one KDE smoothing choice.

Cell 52 – Define the bandwidth robustness question

Type / role	Markdown
Code or content summary	Documented the 0.5×, 1×, 2× bandwidth sensitivity plan and explicitly rejected exhaustive optimization.
Stored output	No computational output.
Interpretation / decision	Set a qualitative robustness goal: does the raw>co-moving conclusion survive reasonable smoothing changes?
Concepts taught / reinforced	Robustness design; scope control.

Cell 53 – Package the raw/co-moving asymmetry calculation

Type / role	Code
Code or content summary	Defined <code>compute_asymmetry_pair()</code> to compute shared baseline bandwidth × multiplier, symmetric grid, two KDEs, KL metrics, and return a two-row <code>DataFrame</code> .
Stored output	No direct output.
Interpretation / decision	Turned the baseline analysis into a reusable robustness primitive.
Concepts taught / reinforced	Reusable analysis functions; <code>DataFrame</code> return; shared hyperparameters.

Cell 54 – Run the tidy bandwidth sensitivity sweep

Type / role	Code
Code or content summary	Looped τ and multipliers [0.5,1,2], called <code>compute_asymmetry_pair</code> , concatenated results.
Stored output	18-row tidy sensitivity table generated.
Interpretation / decision	Prepared systematic comparison without manual copy/paste.
Concepts taught / reinforced	List of <code>DataFrames</code> ; <code>concat</code> ; nested loops.

Cell 55 – Pivot bandwidth results for interpretation

Type / role	Code
Code or content summary	Pivoted to raw/co-moving columns and computed <code>raw_minus_comoving</code> .
Stored output	At every lag and multiplier <code>raw_minus_comoving</code> stayed positive. Example τ 1: 0.345051 at 0.5×, 0.199199 at 1×, 0.108246 at 2×. Absolute l decreased strongly as smoothing increased.
Interpretation / decision	Qualitative ordering was robust, while absolute metric magnitude was bandwidth-dependent. This became a central interpretation caveat.
Concepts taught / reinforced	pivot; wide/long reshaping; qualitative vs numerical robustness.

τ	multiplier	co-moving	raw	raw-co
1	0.5×	1.287617	1.632668	0.345051
1	1×	0.482956	0.682156	0.199199
1	2×	0.159777	0.268023	0.108246
2	0.5×	1.104216	1.379156	0.274940
2	1×	0.437690	0.568221	0.130532
2	2×	0.126871	0.217927	0.091055
4	0.5×	1.423834	1.771585	0.347750

τ	multiplier	co-moving	raw	raw-co
4	1×	0.541848	0.792825	0.250977
4	2×	0.176811	0.302440	0.125630

BANDWIDTH CONCLUSION Absolute KL magnitude is strongly smoothing-dependent, but raw > co-moving is preserved for every tested lag at 0.5×, 1×, and 2× bandwidth. Therefore the robust statement is qualitative ordering, not a uniquely determined absolute number.

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10. Cells 56–59: cell-level bootstrap uncertainty

This block treats cells, rather than individual steps, as the resampling units and quantifies sampling uncertainty in raw, co-moving, and their difference.

Cell 56 – Define cell-level bootstrap uncertainty

Type / role	Markdown
Code or content summary	Explained why within-cell steps are dependent and specified resampling cells with replacement while keeping raw/co-moving pairing.
Stored output	No computational output.
Interpretation / decision	Avoided the naive assumption that all displacement rows are independent replicates.
Concepts taught / reinforced	Cluster bootstrap; dependence; paired resampling.

Cell 57 – Define paired cell bootstrap sampling

Type / role	Code
Code or content summary	Built sets of raw/co cell IDs, intersected, sampled cells with replacement using a NumPy Generator, collected all dx/dy rows for each sampled cell, vstacked points.
Stored output	No direct output.
Interpretation / decision	Preserved within-cell blocks and matched raw/co-moving biological sampling units.
Concepts taught / reinforced	set/intersection; rng.choice; replace=True; list blocks; np.vstack.

Cell 58 – Run 100 cell-bootstrap replicates per lag

Type / role	Code
Code or content summary	Set N_BOOTSTRAP=100 and RANDOM_SEED=2026; for each lag resampled cells, recomputed asymmetry pair on 81×81 grid, stored raw/co/difference.
Stored output	300 bootstrap rows generated (3 lags ×100); reproducible due fixed seed.
Interpretation / decision	Provided exploratory sampling uncertainty without turning the project into high-cost resampling.
Concepts taught / reinforced	Random generators; seeds; Monte Carlo loops; nonlinear statistic recomputation.

Cell 59 – Summarize bootstrap distributions

Type / role	Code
Code or content summary	For each lag and metric raw/co/raw_minus_co, computed mean and 2.5/97.5% quantiles.
Stored output	raw-co intervals: τ_1 [0.041472,0.550427]; τ_2 [-0.082389,0.524471]; τ_4 [-0.055736,0.677328].
Interpretation / decision	The raw>co difference was cell-bootstrap stable at τ_1 but uncertain at τ_2/τ_4 . Bootstrap means exceeded point estimates because the nonlinear estimator is recomputed on resampled cell compositions.
Concepts taught / reinforced	Percentile bootstrap intervals; nonlinear-estimator behavior.

τ	metric	bootstrap mean	2.5%	97.5%
1	raw	0.976148	0.596142	1.501263
1	co-moving	0.728371	0.443056	1.202298

τ	metric	bootstrap mean	2.5%	97.5%
1	raw-co	0.247777	0.041472	0.550427
2	raw	0.821971	0.410549	1.441254
2	co-moving	0.653052	0.383908	1.109211
2	raw-co	0.168919	-0.082389	0.524471
4	raw	1.181971	0.605889	2.008957
4	co-moving	0.918295	0.436351	1.529409
4	raw-co	0.263675	-0.055736	0.677328

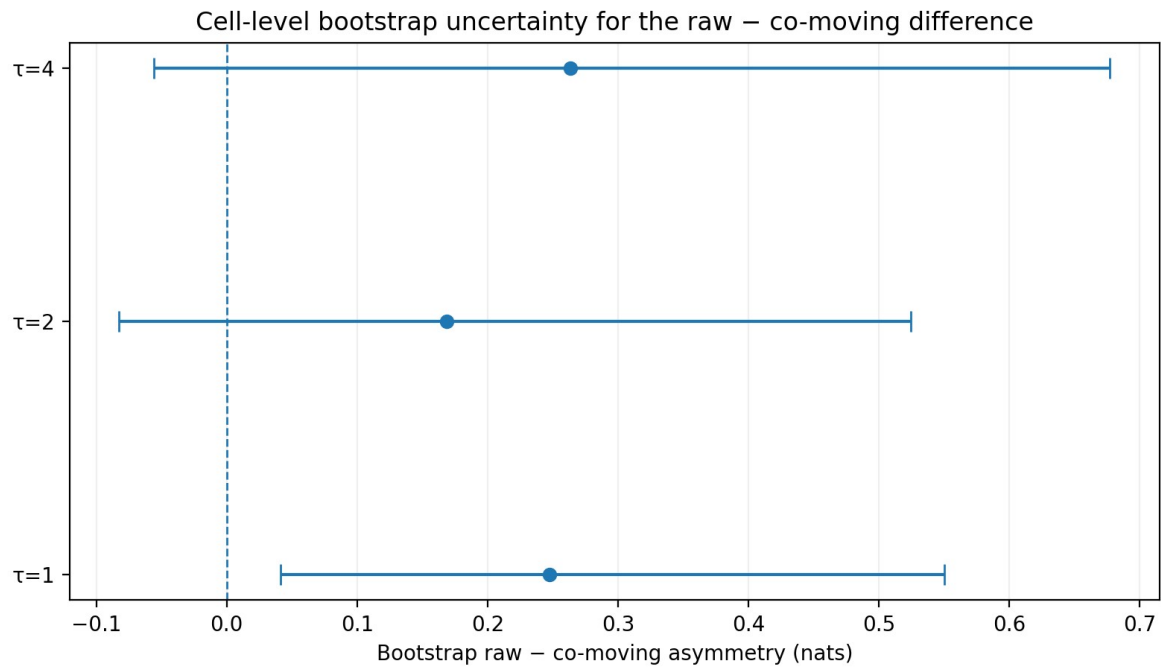


Figure 10. Cell-level bootstrap intervals for raw-co-moving asymmetry. $\tau=1$ remains positive across the exploratory interval; $\tau=2$ and $\tau=4$ intervals cross zero.

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11. Cells 60–62: inversion-symmetrized null control

This block asks how much positive KL asymmetry the finite-sample KDE estimator can produce after systematic inversion preference is deliberately destroyed.

Cell 60 – Define the inversion-symmetrized null control

Type / role	Markdown
Code or content summary	Specified random whole-vector sign flips $\Delta \rightarrow \pm\Delta$ with 50/50 probability; emphasized dx/dy signs are flipped together; framed result as exploratory reference rather than definitive hypothesis test.
Stored output	No computational output.
Interpretation / decision	Created an estimator/background reference that destroys systematic inversion preference while preserving observed magnitudes and much sample structure.
Concepts taught / reinforced	Null model design; vector sign symmetry; conservative inference.

Cell 61 – Simulate the null distributions

Type / role	Code
Code or content summary	Set N_NULL=100 and seed 2027; for each lag/representation generated random ± 1 signs, multiplied whole vectors, reused baseline bandwidth/grid, evaluated null KDE and asymmetry.
Stored output	600 null simulations (3 lags $\times$ 2 representations $\times$ 100).
Interpretation / decision	Estimated how much positive KL asymmetry can appear from finite sampling + KDE even after imposed inversion symmetry.
Concepts taught / reinforced	Random sign arrays; simulation; estimator bias/background.

Cell 62 – Compare observed asymmetry with null mean and q95

Type / role	Code
Code or content summary	Grouped null simulations for mean/q95, merged with observed table, computed observed-minus-null-mean and Boolean above_null_q95.
Stored output	τ_1 raw: 0.682156 > q95 0.538181 (True); τ_1 co: 0.482956 < 0.517069 (False). τ_2 and τ_4 raw and co-moving were all above their null q95.
Interpretation / decision	At τ_1 the co-moving result was not clearly beyond the simple null background; at longer lags both branches were. This was treated as exploratory evidence, not formal $p < 0.05$ proof.
Concepts taught / reinforced	groupby/agg; quantile lambda; merge; Boolean thresholds.

τ	repr	observed	null mean	null q95	obs-null mean	>q95?
1	raw	0.682156	0.420430	0.538181	0.261726	True
1	co	0.482956	0.389444	0.517069	0.093513	False
2	raw	0.568221	0.229393	0.318185	0.338828	True
2	co	0.437690	0.218620	0.288457	0.219070	True
4	raw	0.792825	0.215770	0.319931	0.577055	True
4	co	0.541848	0.229415	0.353751	0.312434	True

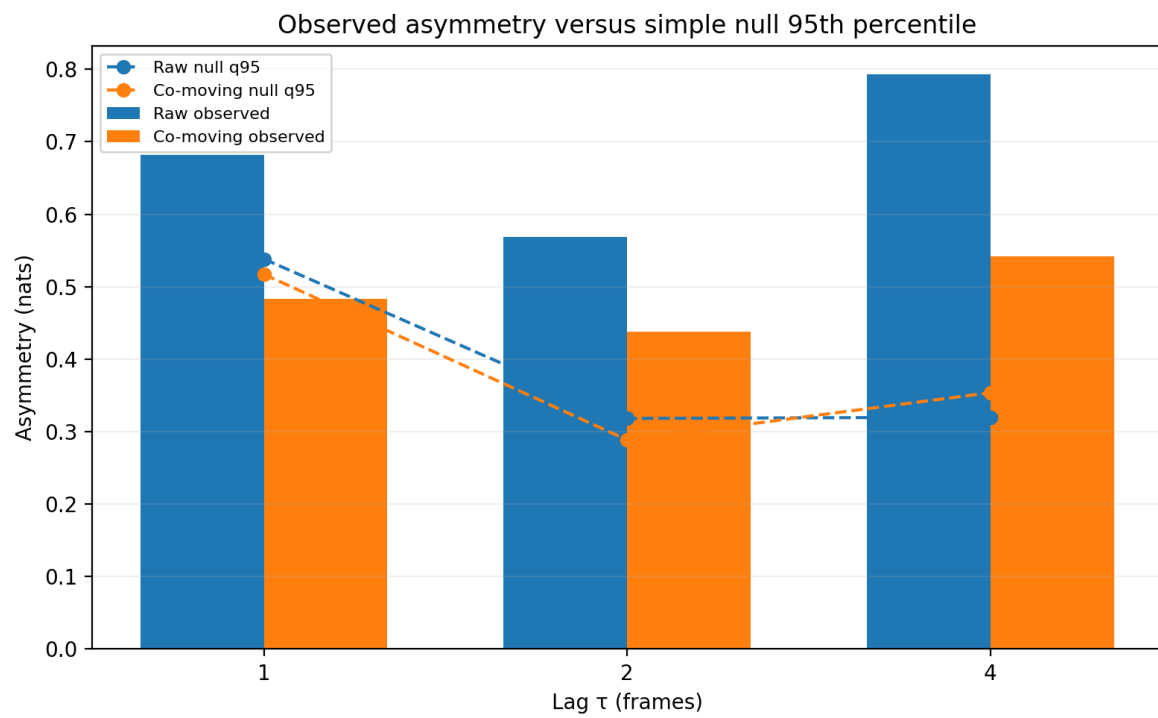


Figure 11. Observed asymmetry versus the 95th percentile of the simple inversion-symmetrized null. Co-moving $\tau=1$ is the only baseline estimate below its branch-specific q95.

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12. Cells 63–67: constant-y sensitivity, synthesis, and final validation

This block fulfills the planned simple data-quality sensitivity analysis, assembles the master table, fixes final language, and closes Phase 5 with validation assertions.

Cell 63 — Create the constant-y exclusion sensitivity dataset

Type / role	Code
Code or content summary	Defined <code>CONSTANT_Y_CELLS=[2,12,14,30,31,26,21]</code> and filtered them out with <code>~isin()</code> .
Stored output	Rows with all cells: 2348; after exclusion: 1986.
Interpretation / decision	Executed the promised simple sensitivity analysis instead of forensic tracking validation.
Concepts taught / reinforced	<code>isin</code> ; Boolean negation <code>~</code> ; sensitivity dataset.

Cell 64 — Recompute asymmetry with and without constant-y tracks

Type / role	Code
Code or content summary	For each lag and representation, computed baseline all-track metric and filtered metric; stored absolute difference and relative change.
Stored output	Relative changes: τ_1 raw -4.98%, co -4.71%; τ_2 raw -16.48%, co -17.78%; τ_4 raw -15.85%, co -19.07%.
Interpretation / decision	Constant-y tracks contributed quantitatively, especially at longer lags, but did not remove the <code>raw>co</code> qualitative pattern.
Concepts taught / reinforced	Sensitivity analysis; relative change.

Cell 65 — Assemble the Phase 5 master results table

Type / role	Code
Code or content summary	Merged baseline asymmetry, null summary, and constant-y sensitivity into <code>phase5_final</code> .
Stored output	Six-row final table containing τ , representation, <code>n_steps</code> , observed <code>I</code> , null mean/q95, above-null flag, filtered <code>I</code> , relative change.
Interpretation / decision	Created the compact results object used for the final conclusion and handoff.
Concepts taught / reinforced	DataFrame merge; synthesis table.

Cell 66 — Write the final Phase 5 scientific conclusion

Type / role	Markdown
Code or content summary	Summarized empirical distributions, FD binning, 1D/2D KDE, inversion metric, bandwidth sensitivity, bootstrap, null, constant-y limitation; stated exact baseline values and interpretation boundaries.
Stored output	No computational output.
Interpretation / decision	Locked the wording: evidence supports single-displacement spatial inversion asymmetry; common translation accounts for part but not necessarily all; no claim of full temporal irreversibility or entropy production.
Concepts taught / reinforced	Scientific synthesis; claim discipline; handoff narrative.

Cell 67 — Run final Phase 5 validation

Type / role	Code
Code or content summary	Asserted lag set, representation set, six final rows, positive metrics, grid mass 0.99–1.01, and positive baseline raw-minus-co across all lags; displayed final table.
Stored output	“Phase 5 validation: PASSED.” Final six-row phase5_final displayed.
Interpretation / decision	Closed the phase with explicit computational invariants before Git/README documentation and the Phase 6 transition.
Concepts taught / reinforced	Final assertions; validation gates; reproducibility checkpoint.

τ	repr	all tracks	without constant-y	difference	relative change
1	raw	0.682156	0.648158	-0.033997	-4.98%
1	co	0.482956	0.460205	-0.022752	-4.71%
2	raw	0.568221	0.474558	-0.093664	-16.48%
2	co	0.437690	0.359868	-0.077821	-17.78%
4	raw	0.792825	0.667167	-0.125658	-15.85%
4	co	0.541848	0.438515	-0.103334	-19.07%

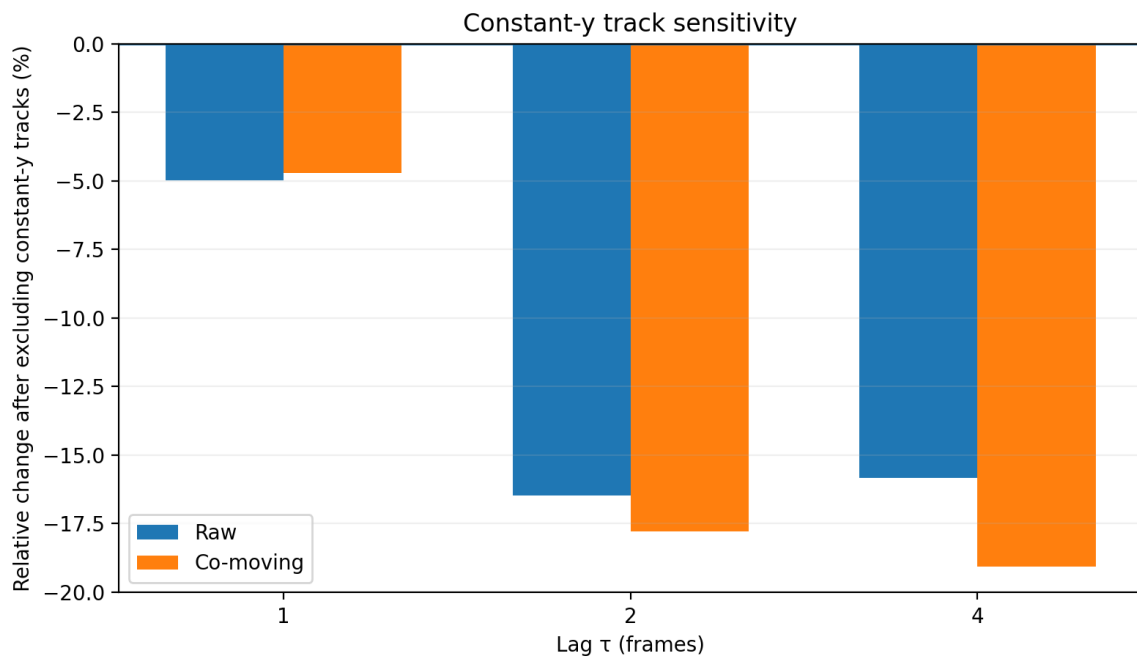


Figure 12. Excluding the seven constant-y tracks lowers the metric but does not reverse the qualitative raw-versus-co-moving ordering.

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13. Integrated Phase 5 results and interpretation

The Phase 5 evidence is internally coherent when the different analyses are read together. Raw trajectories show a stronger inversion asymmetry than co-moving trajectories at every baseline lag. This agrees with Phase 4: the common population translation is a real contributor to directional statistics. However, positive co-moving asymmetry remains, and at $\tau=2$ and $\tau=4$ it exceeds the 95th percentile of the simple inversion-symmetrized null.

Question	Phase 5 answer	Strength / caveat
Does common population translation matter?	Yes. Raw $I_t >$ co-moving I_t at all three lags.	Robust to $0.5\times-2\times$ bandwidth.
Does co-moving asymmetry vanish?	No. Baseline co-moving I_t remains positive.	Positive KL alone is not enough; compare with null.
Does co-moving $\tau=1$ exceed the simple null q95?	No (0.483 vs 0.517).	Evidence at shortest lag weakens after common-motion removal.
Do co-moving $\tau=2$ and $\tau=4$ exceed null q95?	Yes.	Exploratory, not a formal proof because the null is intentionally simple.
Is raw-co difference cell-bootstrap stable?	$\tau=1$ interval positive; $\tau=2/4$ intervals include zero.	Longer-lag difference depends more on sampled cell composition.
Do constant-y tracks dominate the conclusion?	They influence magnitude but do not erase the pattern.	Exclusion lowers I by $\sim 5\%$ at $\tau 1$ and $\sim 16-19\%$ at longer lags.
Is the absolute I_t a physical constant?	No.	It changes strongly with KDE bandwidth.
Did Phase 5 prove time irreversibility?	No.	It measured single-displacement spatial inversion asymmetry.

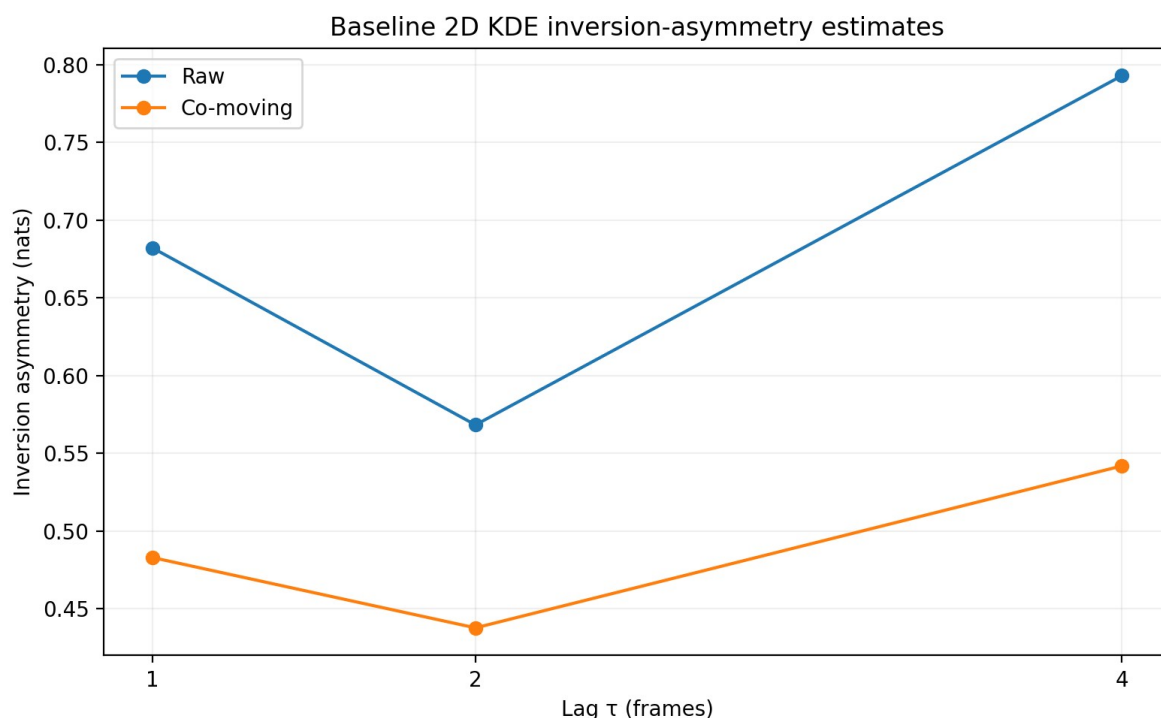


Figure 13. Baseline raw and co-moving inversion-asymmetry values across lag.

BEST DEFENSIBLE PHASE 5 STATEMENT In this exploratory dataset, the estimated displacement distribution is spatially inversion-asymmetric. Removing frame-wise common translation reduces that asymmetry. At longer lags, residual co-moving asymmetry remains above a simple inversion-symmetrized null reference, but the magnitude is estimator- and sample-dependent and is not interpreted as entropy production or full path irreversibility.

14. Decision log, limitations, and the project mindset

Phase 5 contained an explicit course correction in how much diagnostic depth the project should pursue. The project remains rigorous about validity, but it is an exploratory training project rather than a publication-grade tracking-validation study. The decision rule below is now part of the project methodology and should carry into later phases.

Situation	Action rule	Phase 5 example
Problem changes the definition of the observable or invalidates the pipeline	Stop and resolve before continuing.	Exact-lag bug discovered in Phase 4; Phase 5 preflight revalidated exact-frame logic.
Issue can materially bias results but does not invalidate the whole dataset	Document it and run a compact sensitivity analysis.	Seven constant-y tracks; retained in primary analysis, excluded once as sensitivity.
Interesting anomaly with plausible biological explanation and no obvious failure	Inspect enough to understand structure, then retain.	Cell 10 large τ_4 tail traced to sustained/overlapping motion episodes.
Potential publication-level provenance question not needed for exploratory objective	Do not open an indefinite forensic subproject.	No extended Fiji/raw-image boundary investigation for constant-y tracks.
Estimator choice changes absolute values	Test a reasonable robustness range and report dependence.	Bandwidth $0.5\times/1\times/2\times$.
Inference could be overclaimed	Use conservative language and define what is not shown.	KL inversion asymmetry $\neq$ entropy production or full time reversal.

PROJECT MINDSET Scientific and reproducible, but proportionate. We investigate issues that can fundamentally break the analysis; otherwise we disclose limitations, use simple sensitivity analyses, and continue the learning roadmap instead of optimizing every detail to manuscript standards.

Specific unresolved limitations carried forward: (1) several tracks have constant raw y coordinates; cause not resolved, (2) step observations within a cell and overlapping windows are correlated, (3) KDE asymmetry depends on bandwidth, (4) only sequence 01 and selected lags were studied, (5) the null model is deliberately simple, (6) Phase 5 contains notebook-local reusable functions not yet refactored into `src/density.py` and `src/metrics.py`, and (7) thermodynamic interpretation is deferred.

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15. Python and scientific-computing concepts learned in Phase 5

Concept	Phase 5 use
DataFrame / Series	Tables versus one-dimensional labeled vectors; filtering, selection, aggregation.
.loc and Boolean masks	Select rows satisfying representation/lag/cell conditions and selected columns.
Dictionary	Explicit key→value mapping, used for expected counts and component-to-column mapping.
List	Ordered mutable collection used for TAUS, cell IDs, result blocks, and lists of DataFrames.
Tuple unpacking	for component, column_name in dict.items(); paired return values from functions.
Set / intersection	Unique cell IDs and paired raw/co-moving cell bootstrap universe.
Functions	Reusable named computation; parameters, defaults, return values, type hints.
Defensive programming	ValueError for invalid components, insufficient samples, or zero IQR; assertions for invariants.
Nested loops	Systematic enumeration of representation × lag × component or simulation replicate.
List of dictionaries	Build tidy records first, then convert to DataFrame.
pd.concat	Vertical stacking of raw/co-moving rows or result tables.
groupby / agg	Compute per-lag, per-cell, and null summaries.
pivot	Convert tidy sensitivity results into side-by-side raw/co-moving columns.
merge	Combine observed, null, and sensitivity tables by keyed columns.
NumPy arrays	n×2 displacement point matrices for 2D KDE.
reshape / ravel / meshgrid	Adapt 1D arrays for sklearn and construct/evaluate 2D grids.
np.hypot	Euclidean 2D step magnitude $\sqrt{(dx^2+dy^2)}$.
Random Generator + seed	Reproducible bootstrap and null simulation.
KDE API	fit → score_samples(log density) → exp → density.
Assertions	Final computational gates: lags, representations, row count, positive metrics, grid mass, raw-co sign.

A key software-engineering lesson was that exploratory notebook code can be progressively functionized without immediate over-refactoring. In Phase 5, reusable functions were first developed locally because the analysis was still evolving. The plan is to move functions into src/ only when later phases actually reuse them.

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16. Reproducibility, code architecture, Git, and notebook design

Phase 5 was developed on a dedicated branch named phase-5-density, validated, pushed, merged into main, and documented in the README. The current README inspected on 5 August 2026 lists Phase 5 as complete and Phase 6 as next. The project test suite remains 9 passing tests; those tests primarily protect the exact-lag step-construction layer rather than the notebook-local KDE functions.

File / component	Role after Phase 5
notebooks/O5_displacement_density_MSC01.ipynb	Authoritative executed Phase 5 learning/research notebook; stores all intermediate outputs and final validation.
src/steps.py	Authoritative exact-frame displacement construction reused by Phase 5.
src/density.py	Placeholder/reuse target if KDE helpers are needed again.
src/metrics.py	Placeholder/reuse target if asymmetry/sequence metrics are needed again.
README.md	Public high-level summary; states Phase 5 completed and Phase 6 next.
instructions/Phase 5 standalone handbook	This document: deep educational and scientific continuation record.
data/*.csv	Local/ignored biological inputs and derived analysis artifacts; not guaranteed in a fresh clone.
tests/test_steps.py	Regression protection for exact lags, missing frames, duplicates, etc.

Notebook-size decision. Notebook O5 contains more than 60 cells. That is acceptable as a research + teaching notebook because concepts were introduced stepwise and diagnostics were preserved as history. It is not the target style for every future phase. The default from Phase 6 onward is one clearly named notebook per phase when practical, with major reusable logic moved to src/ if it is truly reused; if a phase must split, filenames must explicitly carry the same phase number.

DO NOT RETROACTIVELY REFACTOR NOW The current notebook is a valuable learning record. Refactor only when reuse in Phase 6+ creates a real benefit; do not spend time shortening Notebook O5 merely to meet an arbitrary cell count.

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17. Phase 6 handoff and continuation protocol

Phase 5 tested spatial inversion of single displacement vectors. Phase 6 must add temporal order. For a forward sequence ($\Delta 1, \Delta 2, \Delta 3$), the time-reversed sequence is ($-\Delta 3, -\Delta 2, -\Delta 1$): the vector signs reverse and the temporal order reverses.

Carry forward	Instruction
Raw and co-moving branches	Keep both; Phase 5 shows common translation changes one-step asymmetry.
Exact-lag construction	Do not replace with row shifts. Reuse <code>src/steps.py</code> and preserve exact frame continuity.
Phase 5 It	Keep as a descriptive one-step inversion baseline, not as the final time-irreversibility statistic.
Density functions	Reuse/refactor only if Phase 6 needs them; avoid copying large function blocks blindly.
Resampling unit	Prefer cell-level/cluster-aware uncertainty over individual-step resampling.
Null philosophy	Match the null to the metric; for sequence reversal, sign randomization alone is not a complete temporal null.
Data-quality limitations	Do not reopen constant-y provenance unless a Phase 6 result is qualitatively driven by those tracks. A simple sensitivity analysis is enough by default.
Project scope	Exploratory, educational, reproducible; rigorous about fundamental validity, proportionate about secondary anomalies.

Suggested Phase 6 notebook naming: `notebooks/06_phase6_sequence_irreversibility_MSC01.ipynb`. Default preference: one notebook per phase. If Phase 6 becomes too large, use names such as `06A_phase6_sequence_construction_MSC01.ipynb` and `06B_phase6_sequence_irreversibility_MSC01.ipynb` so the phase relationship remains obvious.

1. Start on main; pull latest; confirm clean status and 9 passing tests.
2. Create a Phase 6 branch only after the Phase 5 handbook/README documentation state is clean.
3. Define exactly what constitutes a contiguous ordered sequence and how missing frames affect eligibility.
4. Build a small hand-checkable toy sequence test before applying to MSC01.
5. Compare forward sequence statistics with the correctly reversed sequence ($-\Delta k, \dots, -\Delta 1$).
6. Carry both raw and co-moving analyses, but avoid repeating Phase 5 diagnostics unless the new metric requires them.
7. Use compact robustness: a few sequence lengths / estimator settings, cell-level uncertainty, and a null suited to sequence order.
8. Keep thermodynamic entropy-production language deferred until theoretical assumptions are explicitly justified.

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Appendix A. Key Phase 5 code listings

The executed notebook remains the authoritative complete cell-by-cell code record. The listings below reproduce the core reusable algorithms that define the Phase 5 analysis, while the main body explains every cell's purpose, output, interpretation, and decision impact.

Exact-lag input baseline

```
EXPECTED_COUNTS = {1: 433, 2: 398, 4: 343}
raw_counts = steps_raw.groupby("tau_frames").size().to_dict()
comoving_counts = steps_comoving.groupby("tau_frames").size().to_dict()
assert raw_counts == EXPECTED_COUNTS
assert comoving_counts == EXPECTED_COUNTS
```

Freedman-Diaconis width

```
def freedman_diaconis_bin_width(values: pd.Series) -> float:
    clean_values = values.dropna()
    n = len(clean_values)
    q25 = clean_values.quantile(0.25)
    q75 = clean_values.quantile(0.75)
    iqr = q75 - q25
    return float(2 * iqr / np.cbrt(n))
```

1D KDE evaluation

```
def evaluate_kde_1d(values, grid, bandwidth):
    sample = values.dropna().to_numpy().reshape(-1, 1)
    grid_2d = grid.reshape(-1, 1)
    kde = KernelDensity(kernel="gaussian", bandwidth=bandwidth)
    kde.fit(sample)
    log_density = kde.score_samples(grid_2d)
    return np.exp(log_density)
```

2D displacement extraction

```
def get_displacement_points(data, representation, tau_frames):
    mask = (data["representation"].eq(representation)
            & data["tau_frames"].eq(tau_frames))
    return (data.loc[mask, ["dx_um", "dy_um"]]
            .dropna().to_numpy())
```

2D KDE evaluation

```
def evaluate_kde_2d(points, x_grid, y_grid, bandwidth):
    xx, yy = np.meshgrid(x_grid, y_grid)
    evaluation_points = np.column_stack([xx.ravel(), yy.ravel()])
    kde = KernelDensity(kernel="gaussian", bandwidth=bandwidth)
    kde.fit(points)
    log_density = kde.score_samples(evaluation_points)
    density = np.exp(log_density).reshape(xx.shape)
    return xx, yy, density
```

Inversion asymmetry

```
def inversion_asymmetry_index(density, x_grid, y_grid):
    dx_grid = x_grid[1] - x_grid[0]
    dy_grid = y_grid[1] - y_grid[0]
    cell_area = dx_grid * dy_grid
    grid_mass = density.sum() * cell_area
    p = density / grid_mass
    p_reversed = p[::-1, ::-1]
    epsilon = p.max() * 1e-12
    p_safe = p + epsilon
    q_safe = p_reversed + epsilon
    p_safe /= p_safe.sum() * cell_area
    q_safe /= q_safe.sum() * cell_area
    log_ratio = np.log(p_safe / q_safe)
```

```
asymmetry = (p_safe * log_ratio).sum() * cell_area
return float(asymmetry), float(grid_mass)
```

Cell-level bootstrap core

```
sampled_cells = rng.choice(common_cells, size=len(common_cells), replace=True)
raw_blocks, comoving_blocks = [], []
for cell_id in sampled_cells:
    raw_blocks.append(raw_tau.loc[raw_tau["cell_id"].eq(cell_id), ["dx_um", "dy_um"]].to_numpy())
    comoving_blocks.append(comoving_tau.loc[comoving_tau["cell_id"].eq(cell_id),
["dx_um", "dy_um"]].to_numpy())
raw_points = np.vstack(raw_blocks)
comoving_points = np.vstack(comoving_blocks)
```

Inversion-symmetrized null

```
signs = null_rng.choice([-1.0, 1.0], size=(len(points), 1))
null_points = points * signs
# The whole 2D vector is multiplied by one sign, so dx and dy invert together.
```

Constant-y sensitivity filter

```
CONSTANT_Y_CELLS = [2, 12, 14, 30, 31, 26, 21]
steps_without_constant_y = steps_analysis.loc[
    ~steps_analysis["cell_id"].isin(CONSTANT_Y_CELLS)
].copy()
```

Final validation

```
assert set(phase5_final["tau_frames"]) == {1, 2, 4}
assert set(phase5_final["representation"]) == {"raw", "comoving"}
assert len(phase5_final) == 6
assert (phase5_final["inversion_asymmetry_nats"] > 0).all()
assert asymmetry_summary["grid_mass"].between(0.99, 1.01).all()
assert (baseline_bandwidth_rows["raw_minus_comoving"] > 0).all()
print("Phase 5 validation: PASSED")
```

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Appendix B. Numerical output ledger

Output family	Verified values / conclusion
Input trajectory shapes	Raw (474,5), co-moving (474,5); 35 cells each; structure identical=True.
Exact step counts	$\tau_1=433$, $\tau_2=398$, $\tau_4=343$; 1174 per branch; combined 2348 rows.
τ_1 dx ranges	Raw -12.8823...7.8000; co-moving -11.3520...7.8261 μm .
Exact zeros (raw τ_1)	$dx=17$; $dy=87$; both=13.
Constant-y cells	[2,12,14,30,31,26,21].
τ_1 max 2D magnitude	12.9816169932 μm ; 9 steps >8 μm .
FD bin counts	Raw: dx/dy τ_1 37/45, τ_2 27/35, τ_4 22/33; co: 37/51, 29/37, 22/37.
τ_1 KDE bandwidths	dx raw/co 0.427429/0.395657; dy raw/co 0.356959/0.316772 μm .
2D baseline bandwidths	τ_1 0.515968; τ_2 0.955681; τ_4 1.755631 μm .
Baseline l_t raw	τ_1 0.682156; τ_2 0.568221; τ_4 0.792825 nats.
Baseline l_t co-moving	τ_1 0.482956; τ_2 0.437690; τ_4 0.541848 nats.
Grid mass	All baseline estimates 0.999996–0.999999.
Bootstrap raw-co CI	τ_1 [0.041472,0.550427]; τ_2 [-0.082389,0.524471]; τ_4 [-0.055736,0.677328].
Null q95 flags	τ_1 raw True, co False; τ_2 raw/co True; τ_4 raw/co True.
Constant-y exclusion	l reduced ~5% at τ_1 and ~16–19% at τ_2/τ_4 ; qualitative pattern retained.
Final validation	Phase 5 validation: PASSED.

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Appendix C. Cell index 1–67

Cell	Short purpose
1	Define Phase 5 scope and stop premature irreversibility claims
2	Import project helpers and define canonical paths
3	Load and validate raw/co-moving trajectories
4	Prove structural identity between representations
5	Recompute exact-lag steps from trajectories
6	Validate the authoritative lag counts
7	Build a tidy raw + co-moving analysis table
8	Compute the first numerical distribution summary
9	Import numerical plotting tools
10	Isolate the first teaching example: $\tau=1$, dx
11	Create shared pedagogical histogram bins
12	Plot the first normalized raw histogram
13	Overlay raw and co-moving $\tau=1$ dx histograms
14	Define reusable displacement selection
15	Define shared-bin helper
16	Define reusable histogram plotting function
17	Generate all six raw/co-moving histogram comparisons
18	Build full numerical distribution diagnostics
19	Display a compact diagnostic table
20	Locate the largest raw $\tau=4$ dy displacements
21	Inspect Cell 10 raw trajectory table
22	Plot Cell 10 raw y-position over time
23	Inspect Cell 10 exact $\tau=1$ steps and 2D magnitudes
24	Plot Cell 10 one-frame dy sequence
25	Count exact-zero dy by cell
26	Separate zero- dy from full zero displacement
27	Quantify large $\tau=1$ 2D step magnitudes
28	Inspect Cell 21, a constant-y example
29	Summarize all 100%-zero- dy tracks
30	Visualize Cell 21 x and y coordinates over time
31	Record the Phase 5 data-quality policy
32	Define Freedman–Diaconis bin-width function
33	Convert FD width into shared raw/co-moving bin edges
34	Audit FD recommendations across all 12 distributions
35	Define FD histogram comparison plotter
36	Generate six FD histogram comparisons
37	Import scikit-learn KDE estimator
38	Define robust 1D Silverman-type bandwidth
39	Fit/evaluate 1D Gaussian KDE
40	Define raw/co-moving KDE comparison plotter
41	Run first KDE checks at $\tau=1$
42	Generate the complete set of 1D KDE comparisons
43	Define the 2D joint-density and inversion problem

Cell	Short purpose
44	Extract 2D displacement point arrays
45	Define robust characteristic scale per coordinate
46	Define a shared isotropic 2D KDE bandwidth
47	Construct symmetric 2D evaluation grids
48	Evaluate a 2D Gaussian KDE on the grid
49	Define the inversion-asymmetry index
50	Compute baseline asymmetry for all lags and branches
51	Visualize raw $\tau=1$ 2D displacement KDE
52	Define the bandwidth robustness question
53	Package the raw/co-moving asymmetry calculation
54	Run the tidy bandwidth sensitivity sweep
55	Pivot bandwidth results for interpretation
56	Define cell-level bootstrap uncertainty
57	Define paired cell bootstrap sampling
58	Run 100 cell-bootstrap replicates per lag
59	Summarize bootstrap distributions
60	Define the inversion-symmetrized null control
61	Simulate the null distributions
62	Compare observed asymmetry with null mean and q95
63	Create the constant-y exclusion sensitivity dataset
64	Recompute asymmetry with and without constant-y tracks
65	Assemble the Phase 5 master results table
66	Write the final Phase 5 scientific conclusion
67	Run final Phase 5 validation

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Appendix D. Glossary and notation

Term	Meaning in this project
$\Delta \tau$	Displacement vector over exact lag τ : $r(t+\tau)-r(t)$.
Raw coordinates	Original TrackMate-derived positions in the laboratory/image coordinate frame.
Co-moving coordinates	Positions after subtracting cumulative frame-wise median common translation.
Random variable	Quantity whose observed values form a sample distribution; e.g., $\Delta X \tau$.
Empirical distribution	Distribution represented by observed finite samples.
Probability density	Function/estimate whose integral over a region gives probability mass.
Histogram	Piecewise-constant empirical density estimator controlled by bin edges.
IQR	Interquartile range $Q75-Q25$; robust spread of the middle 50%.
Freedman–Diaconis	Histogram bin-width rule $h=2 \cdot \text{IQR}/n^{1/3}$.
KDE	Kernel density estimation; smooth density estimate formed by kernels around samples.
Bandwidth	KDE smoothing scale; small=rough/noisy, large=oversmoothed.
Joint density	Density over multiple variables simultaneously; here $p(dx,dy)$.
Spatial inversion	Vector map $(dx,dy) \rightarrow (-dx,-dy)$.
Symmetric grid	Numerical evaluation domain containing matched Δ and $-\Delta$ coordinates.
Grid mass	Numerical integral of KDE over finite grid before renormalization.
KL divergence	$DKL(p q) = \int p \log(p/q)$; nonnegative asymmetry/difference measure in nats with natural log.
Inversion asymmetry I_t	Phase 5 KL divergence between $p\tau(\Delta)$ and $p\tau(-\Delta)$.
nats	Information unit when logarithms use base e .
Cell-level bootstrap	Resampling whole cell IDs with replacement, retaining all selected-cell steps.
Null control	Reference analysis designed to remove the target structure; here random whole-vector $\pm$ sign symmetrization.
Sensitivity analysis	Repeat analysis under a plausible alternative data/parameter choice to assess dependence.
Constant-y track	Track whose stored raw y coordinate is identical across all its observations.
Overlapping lag windows	Multi-frame displacement windows sharing underlying one-frame intervals; not independent events.
Temporal irreversibility	Difference between probabilities of ordered forward paths and correctly time-reversed paths; not established by Phase 5.
Entropy production	Thermodynamic quantity requiring additional theory/assumptions; not estimated as a validated physical value in Phase 5.

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Appendix E. New-chat handoff prompt

Paste the Phase 0–4 Master Handbook and this Phase 5 handbook into the new chat, then use the prompt below. It is intentionally explicit so the next chat does not reopen settled work or silently strengthen claims.

We are continuing the Cell Motility Irreversibility project. Treat the attached Phase 0–4 Master Handbook and the standalone Phase 5 Handbook as the authoritative project record.

Current validated state:

- Phase 5 is complete and validated.
- Exact-lag steps remain $\tau=1:433$, $\tau=2:398$, $\tau=4:343$ for both raw and co-moving branches.
- Notebook 05 implements empirical histograms, Freedman–Diaconis binning, 1D/2D KDE, $p(\Delta)$ vs $p(-\Delta)$, KL inversion asymmetry, bandwidth sensitivity, cell-level bootstrap, a simple inversion-symmetrized null, constant-y sensitivity, and final validation.
- Baseline raw $I\tau = 0.682156, 0.568221, 0.792825$; co-moving $I\tau = 0.482956, 0.437690, 0.541848$ for $\tau=1,2,4$.
- Raw > co-moving is robust across $0.5\times/1\times/2\times$ bandwidth, but absolute I is bandwidth-dependent.
- Cell-bootstrap raw-co difference is clearly positive at $\tau=1$ but intervals cross zero at $\tau=2$ and $\tau=4$.
- Simple null q95: raw $\tau 1/2/4$ above; co-moving $\tau 1$ below, $\tau 2/4$ above.
- Seven constant-y tracks affect metric magnitude but do not remove the qualitative pattern; do not reopen forensic provenance by default.
- Interpretation is single-displacement spatial inversion asymmetry, NOT full time irreversibility and NOT entropy production.

Project mindset:

This is a serious exploratory and educational research project, not a publication-grade validation study. Investigate problems deeply only when they can fundamentally invalidate the observable or pipeline. For smaller data limitations, document them and use a simple sensitivity analysis. Avoid opening long forensic side projects unless the main conclusion depends on them. Continue teaching Python, mathematics/statistics, biophysics, reproducibility, Git/GitHub, and scientific reasoning step-by-step, but keep notebook architecture more compact than Phase 5 when practical. Default preference is one clearly phase-labelled notebook per phase.

Next task:

Start Phase 6: ordered multi-step sequence-level time reversal. First explain the conceptual difference between spatial inversion and true time reversal. Then design a compact Phase 6 roadmap and notebook, including continuity rules, forward/reversed sequence construction, a toy hand-checkable test, raw/co-moving comparison, an appropriate sequence-level metric, cell-level uncertainty, and a null suited to temporal order. Do not call any result entropy production unless its theoretical assumptions are explicitly justified.

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